



principles of Microeconomics

MICROECONOMICS (University of Nairobi)

MICROECONOMICS NOTES (EC101)

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LEVEL: FIRST SEMESTER OF FIRST LEVEL

CHAPTER 1 : FUNDAMENTAL CONCEPTS OF ECONOMICS

ECONOMICS is the study of the ways that individuals and societies allocate their limited resources to try to satisfy their unlimited wants. The major task of economics is studying and evaluating alternatives.

ECONOMICS is a decision science concerned with the choices we make and the consequences of those choices for others and ourselves. In fact, the central forms of economics are on choice and decision-making.

ECONOMICS is also a behavioral and historical science, drawing upon and extending the research of psychologists, anthropologists, sociologists and historians. Moreover, economics is a reflective and moral science often involving the study of problems that puzzle legal scholars, political scientists and philosophers.

MACROECONOMICS is the study of very large, economy wide aggregate variables such as various indicators of the levels of total economic activity. Thus macroeconomic analysis is concerned with things like the banking and monetary systems, and how the levels of Gross National Product (GNP), National Income (NI), unemployment, inflation and economic growth are determined. Macroeconomics also considers such things as the effect of broad government policies, including total government spending and the rates and levels of taxes or rates of growth in the supply of money.

MICROECONOMICS is concerned with individual decision-making; the allocation of resources; and how prices, production, and the distribution of income are determined. It focuses upon the individual and interactive behaviors of households, firms, and fairly specific governmental units. Thus microeconomics emphasizes the composition of economic activity and hence components of our economic system.

All contemporary economists agree that both macroeconomics and microeconomics are essential. An understanding of both is necessary for an accurate perception of how the economy operates.

Scarcity

Humans have many different types of wants and needs. Economics looks only at man's material wants and needs. These are satisfied by *consuming* (using) either *goods* (physical items such as food) or *services* (non-physical items such as heating).

Limited Resources

Commodities (goods and services) are produced by using *resources*. The resources shown in Table 1.1 are sometimes called *factors of production*.

Table 1.1 Different types of resources

Type	Description	Reward
Land	All natural resources	Rent
Labour	The physical and mental work of people	Wages
Capital	All man-made tools and machines	Interest
Enterprise	All managers and organisers	Profit

Types of Commodities

A *free good* is available without the use of resources. There is zero opportunity cost, for example air.
An *economic good* is a commodity in limited supply.

Expenditure on *producer* or *capital goods* is called *investment*.

The Economic Problem

The *economic problem* refers to the *scarcity* of commodities. There is only a limited amount of resources available to produce the unlimited amount of goods and services we desire.

Society has to decide which commodities to make. For example, do we make missiles or hospitals? We have to decide how to make those commodities. Do we employ robot arms or workers? Who is going to use the goods that are eventually made?

Opportunity Cost

The *opportunity cost* principle states the cost of one good in terms of the next best alternative. For example, a gardener may decide to grow carrots. The opportunity cost of his carrot harvest is the alternative crop that might have been grown instead (e.g. potatoes).

Table 1.2 Examples of opportunity cost decisions

<u>Group</u>	<u>Decision</u>
Individual	Should I buy a record or a revision book?
School	Should we build a music block or tennis courts?
Country	Should we increase police pay or pensions?

Economic Systems

An *economic system* is the way a society sets about *allocating* (deciding) which goods to produce and in which quantities. Different countries have different methods of tackling the economic problem. There are three main types of economic systems.

Market Economies

A *market* or *capitalist economy* is where resources are allocated by prices without government intervention. The USA and Hong Kong are examples of market economies where firms decide the type and quantity of goods to be made in response to consumers needs. An increase in the price of one

good encourages producers to switch resources into the production of that commodity. Consumers decide the type and quantity of goods to be bought. A decrease in the price of one good encourages consumers to switch to buying that commodity. People on high incomes are able to buy more goods and services than are the less well off.

Command Economies

In a *command-planned* or *socialist* economy the government owns most resources and decides on the type and quantity of a good to be made. The USSR and North Korea are examples of command economies. The government sets output targets for each district and factory and allocates the necessary resources. Incomes are often more evenly spread out than in other types of economy.

Mixed Economies

In a *mixed economy* privately owned firms generally produce goods while the government organises the manufacture of essential goods and services such as education, health care, energy and communication. The Zimbabwean economy is a good example of a mixed economy because it has both privately owned firms and parastatals (that produce goods and services that are perceived to be of strategic importance to the economy).

POSITIVE AND NORMATIVE STATEMENTS

Positive statements concern what is, was, or will be; they assert alleged facts about the universe in which we live.

Normative statements concern what ought to be; they depend on our value judgements about what is good or bad. As such they are inextricably bound up with our philosophical, cultural, and religious positions.

To illustrate the distinction, consider some assertions, questions, and hypotheses that can be classified as positive or normative. The statement “It is impossible to break up atoms” is a positive one which can quite definitely be (and of course has been) refuted by empirical experimentation; while the statement “scientists ought not to break up atoms” is a normative statement which involves ethical judgments, and cannot be proved right or wrong by evidence.

In economics the questions “what policies will reduce unemployment?” and “what policies will prevent inflation?” are positive ones while the question “ought we to be more concerned about unemployment than about inflation?” is a normative one.

Positive statements such as the one just considered assert things about the world. If it is possible for a statement to be proved wrong by empirical evidence, we call it a **TESTABLE STATEMENT**. Many positive statements are testable and disagreements over them are appropriately handled by an appeal to the facts.

In contrast to positive statements, that are often testable, normative statements are **NEVER** testable. Disagreements over such normative statements as “it is wrong to show excessive violence on TV” or “it is immoral for someone to have sexual relations with another person of the same sex” cannot be settled by an appeal to empirical observations. Normative questions can be discussed rationally, but doing so requires techniques that differ from those required for rational decisions on positive questions. For this reason, it’s convenient to separate normative from positive enquiries. This is done not because the former is less important than the latter, but merely because they must be investigated by different methods.

PRODUCTION POSSIBILITIES

Competitive choices are alternative choices – you have one alternative or another, but not both. These choices may be divisible; you may be able to select more of one good, but you will then receive less of another. Something must give way.

The Production Possibility Frontier

Society as a whole must make choices about the commodities it produces and consumes. We represent this limitation on what the economy can produce by a simple device called the production possibilities frontier. A production possibilities frontier is one of the simplest models of an economy. Several simplifying assumptions underlie this model of the production possibilities frontier.

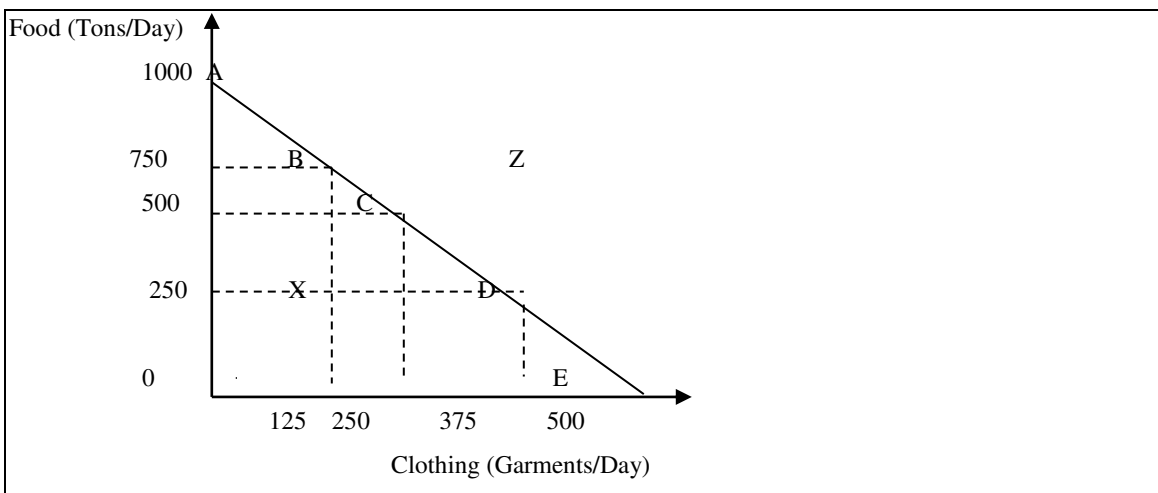
These are:

- a) The factors of production (land, labor, capital and entrepreneurship) are fixed in total supply. However, these resources can be allocated among different types of production.
- b) Technology is constant
- c) The economy operates efficiently, and all resources are fully employed.
- d) Only two goods (products) are produced, that is, food and clothing in our example.

PP SCHEDULE FOR A HYPOTHETICAL ECONOMY

Production Possibility	Food (Tons/Day)	Clothing (Garments/Day)
A	1000	0
B	750	125
C	500	250
D	250	375
E	0	500

PPF FOR THE HYPOTHETICAL ECONOMY



The PPF for an economy illustrates the numerous combinations of food and clothing that can be produced. Combinations A-E from the table above are plotted and connected to obtain the production possibilities frontier. Point X represents underemployment while point Z is unattainable.

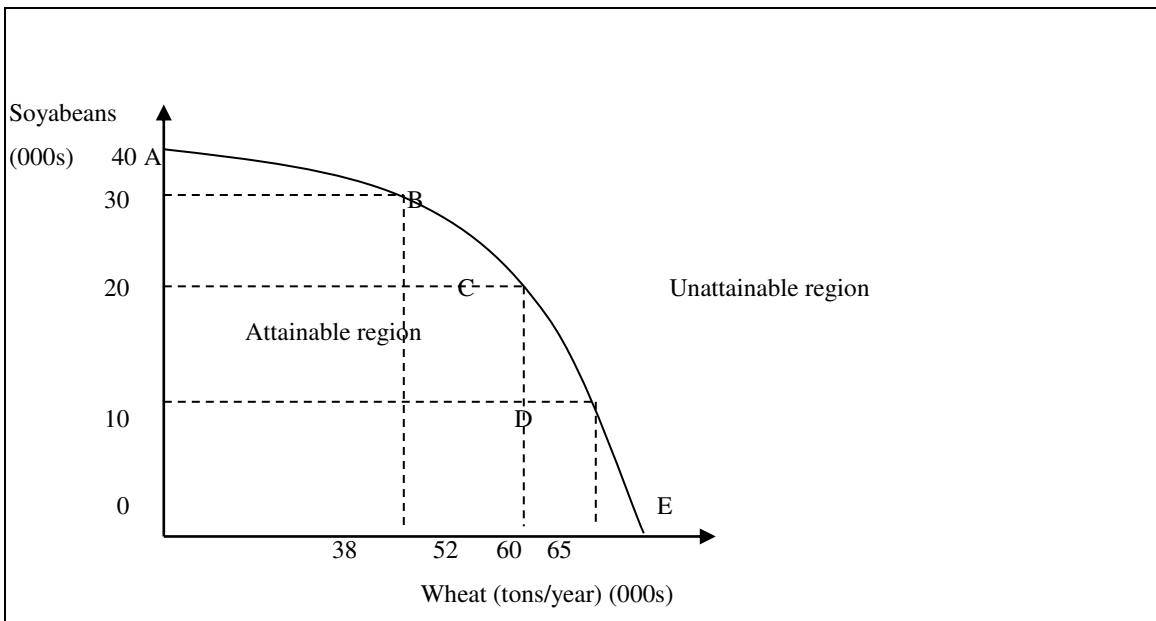
The cost of any decision is its opportunity cost, that is, the value of the next best alternative that the decision forces one to give up. Rational decision making, be it in industry, government, or households, must be based on opportunity cost calculations.

The opportunity cost of any decision is the value of the next alternative that the decision forces the decision-maker to forgo.

PRODUCTION POSSIBILITIES OPEN TO A FARMER

Tons of soyabeans	Tons of wheat	Label
50 000	0	A
40 000	40 000	B
21 000	50 000	C
12 000	60 000	D
0	70 000	E

Production possibilities Frontier for production by a single firm



With a given set of inputs the firm can produce only those output combinations given by points below the PPF. The PPF A-E is not a straight line but one that curves more and more as it nears the axes. That is when the firm specializes in only one product those inputs that are especially adapted to the production of the other good lose at least part of their productivity.

We can see that the slope of the PPF graphically represents the concept of OPPORTUNITY COST. Between points C and B for example the opportunity cost of acquiring 10000 additional tons of soyabeans is 14000 tons of forgone wheat; between B and A the opportunity cost of an additional 10000 tons of soyabeans is 38000 tons of forgone wheat. In general as we move upward to the left along the PPF (toward more soyabeans and less wheat), the opportunity cost of soyabean in terms of

wheat increase. As we move down to the right the opportunity cost of acquiring wheat by giving up soyabeans increases as well.

SCARCITY AND CHOICE FOR THE ENTIRE SOCIETY

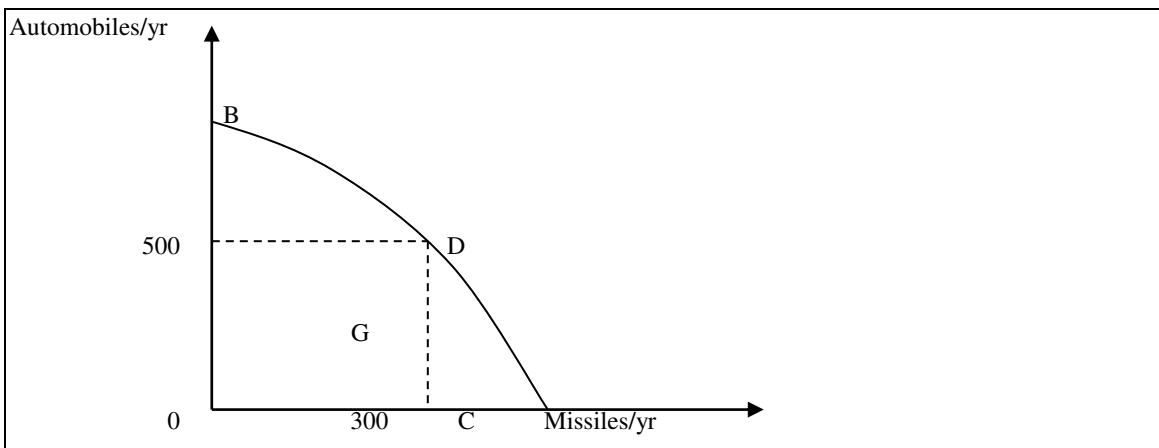
Like an individual firm the entire economy is also constrained by its resources and technology. If the society wants more tractors it will have to give up some luxury cars. If it wants to build more factories and stores it will have to build fewer homes and sports arenas.

The position and shape of the PPF that constraints the choices of the economy are determined by the economy's physical resources, its skills and technology, its willingness to work, and how it has devoted in the past to the construction of factories, research and innovation.

Since the debate over reducing military strength has been so much on the agenda of several nations recently, let us illustrate the nature of society's choices by an example of choosing between military might (represented by missiles) and civilian consumption (represented by automobiles).

PPF FOR THE ENTIRE ECONOMY

The PPF is curved because resources are not perfectly transferable from automobile production to missile production. The limits on available resources place a ceiling, C, on the output of one product and a different ceiling, B, on the output of the other product.



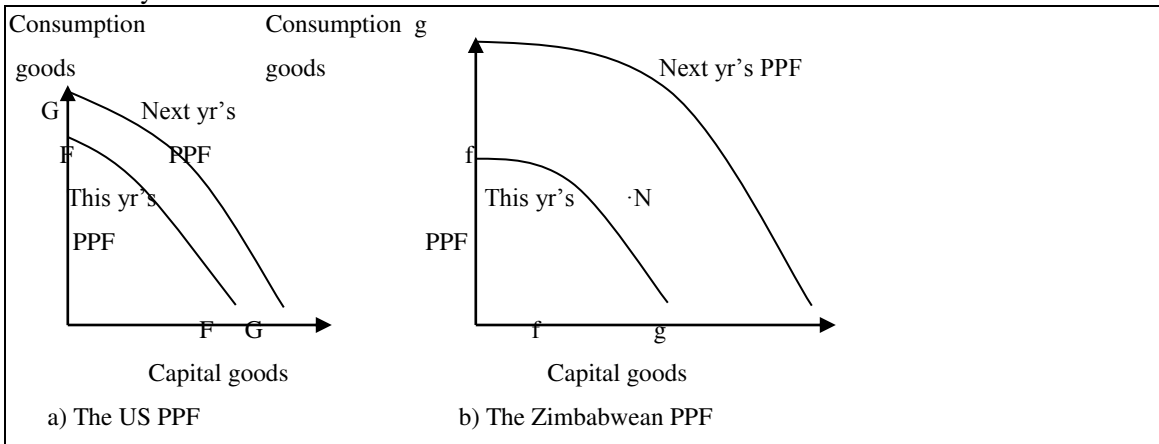
The downward slope of society's PPF implies that hard choices must be made. Our civilian consumption (automobiles) can be increased by decreasing military expenditure and not by rhetoric or by wishing it so. The curvature of the PPF implies that as defense spending increases, it becomes progressively more expensive to "buy" additional military strength (missiles) by sacrificing civilian consumption.

APPLICATION: GROWTH IN THE USA AND ZIMBABWE

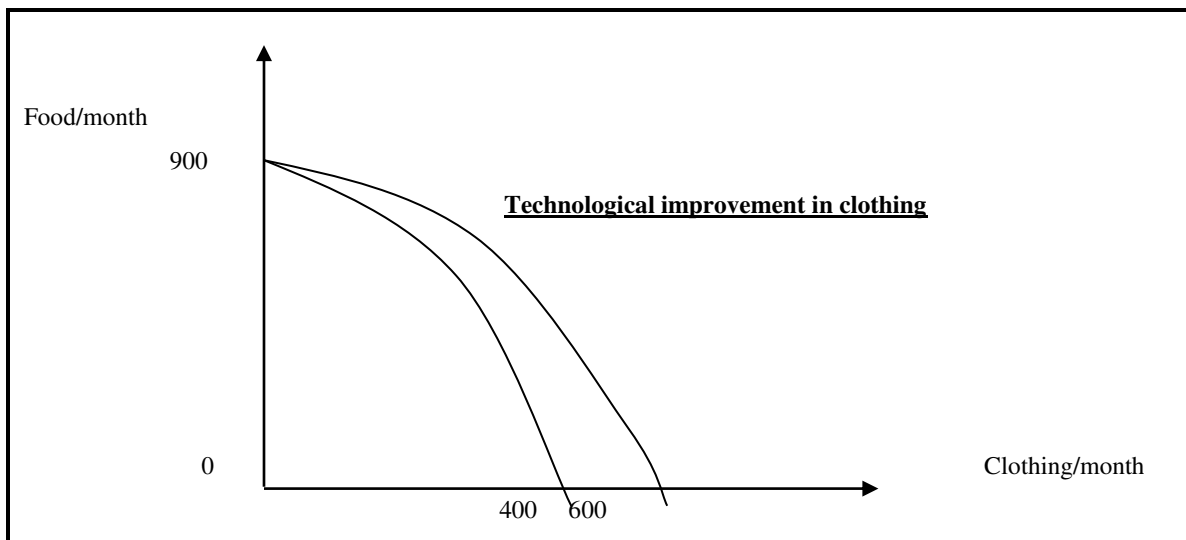
In diagrammatic terms economic growth means that the economy's PPF shifts outward overtime – like the move from FF to GG in figure (a). Why? Because such a shift means that the economy can produce more of both goods shown in the graph. Thus, in the figure, after growth has occurred, it is possible to produce the combination of products represented by points like N. Before growth had occurred point N was beyond the economy's means because it was outside the PPF.

Growth shifts the PPFs FF and ff outward to the frontiers GG and gg, meaning the economies can produce more of both goods than they could before. If the shift in both economies occurs in the same

period of time, then the Zimbabwean economy (panel b) is growing faster than the US economy, (panel a) because the outward shift in (b) is much greater than the one in (a). It is possible for the Zimbabwean economy to grow faster than the US economy because the Zimbabwean economy does not operate at its full capacity while the US economy operates at near or full employment level. If the current land reform in Zimbabwe succeeds then the Zimbabwean economy may grow faster than the US economy in future.



TECHNOLOGICAL IMPROVEMENT IN ONE OF THE COMMODITIES



The entire PPF clearly expands when available resources increase. Note that technological improvement even if only in one commodity expands production opportunities for both commodities because technological advances make resources available for other uses.

CHAPTER 2: MARKET INTERACTION

Definitions

Utility

Utility is the satisfaction people get from *consuming* (using) a good or a service. Utility varies from person to person. Some people get more satisfaction from eating chips than others. Even the same person can gain greater satisfaction by eating chips when hungry than when he has lost his appetite.

MARKET

A market is an institution that brings together buyers and sellers of a commodity. It is usually, but not necessarily, a place where commodities are traded. The market could be:

- Labour market
- Goods market
- Forex market
- Stock market

The market consists of the demand side (driven by buyers) and the supply-side (driven by sellers), interact with the ultimate objective of striking deals and transacting.

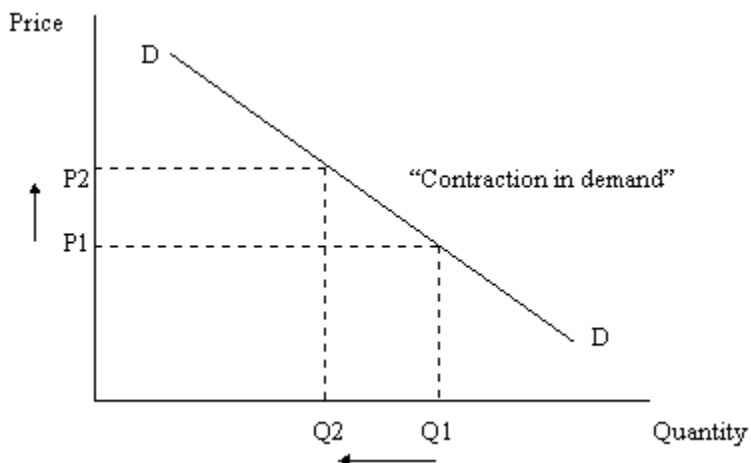
DEMAND

Demand is the amount of a good that consumers are willing and able to buy at a given price. Willingness refers to people's *desire* to own a good. Demand is only referred to as 'effective demand' when there both willingness and ability to acquire a certain commodity. if in Zimbabwe, thousands of people wish to visit Philadelphia, but none can afford it, then the demand for Air Travel to Philadelphia would be zero. Similarly, if in Philadelphia many people can afford to travel to Zimbabwe, but none of them is willing to do so, again we say there is demand for travel along that direction.

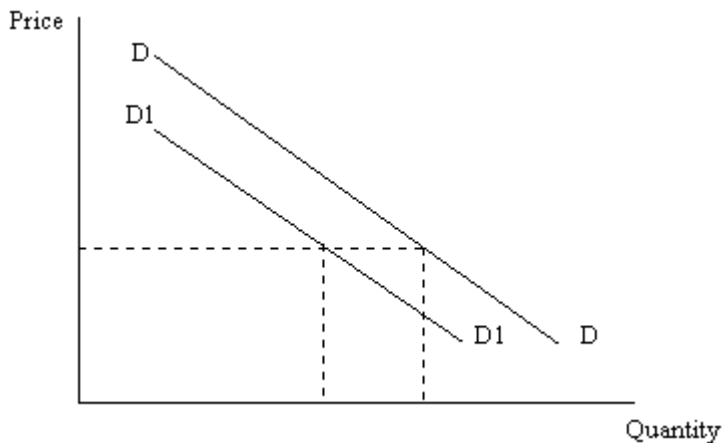
The law of demand says that there is an inverse negative relationship between the price of the commodity and quantity demanded and it is represented by a downward sloping curve. The demand curve is thus drawn based on the law of demand.

Movements Along and Shifts in Demand Curves

A change in price of the commodity itself *never* shifts the demand curve for that good. In the figure below an increase in price results in a *movement* up the demand curve. The fall in the quantity demanded from Q_1 to Q_2 is sometimes called a *contraction* in demand.



A demand curve *shifts* only if there is a change in non-price determinants of demand. In the diagram below a decrease in demand has shifted the demand curve to the left. The new demand curve is D1.



Factors affecting the demand curve:

- Income (Y)
- The price of the good (P)
- Taste of current fashions (T)
- Price of substitute goods (P_s)
- Price complimentary goods (P_c)
- Expectation of change in prices (E_p)
- Size of the population (POP)
- Level of advertising of the good (A)
- Level of advertising of complementary goods (A_c)
- Level of advertising of substitute goods (A_s)

$$Q = f(P, Y, P_s, P_c, E_p, POP, A, A_c, A_s)$$

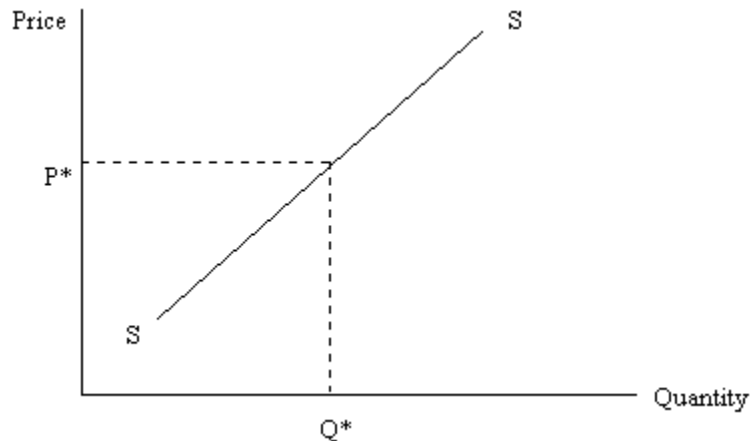
$\begin{matrix} - & + & + & - & +/- & +/- & + & + & - \\ \text{P} & \text{Y} & \text{P}_s & \text{P}_c & \text{E}_p & \text{POP} & \text{A} & \text{A}_c & \text{A}_s \end{matrix}$

A change in any one of the factors in the list above, except P , will shift the demand curve. The signs above each of the variables indicates the direction of the relationship.

SUPPLY

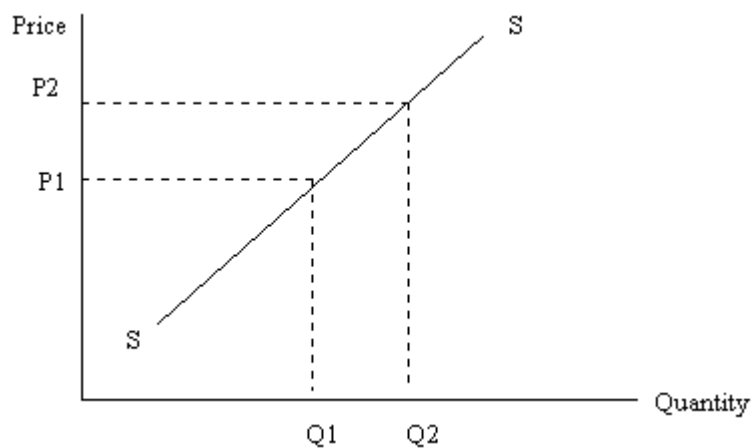
The law of supply states that there is a positive relationship between the price of the good and the quantity supplied. The supply curve is represented by an upward sloping curve.

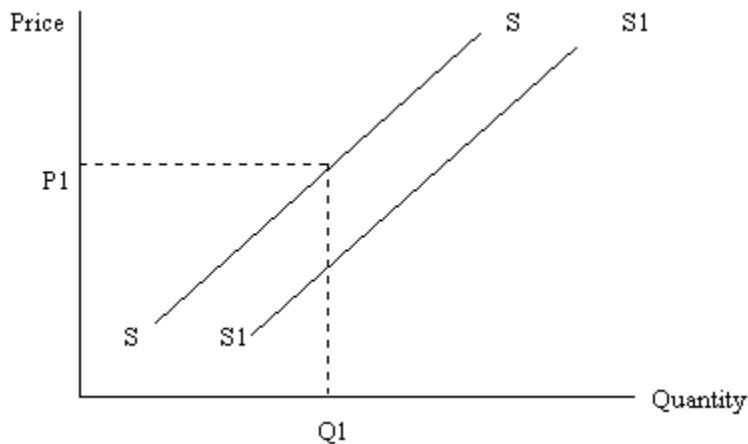
The *supply curve* labelled SS in the figure below shows the amount of a good one or more producers are prepared to sell at different prices.



Movements Along and Shifts in Supply Curves

A change in price *never* shifts the supply curve for that good. In the diagram below an increase in price results in a *movement* up the supply curve. The increase in quantity supplied from Q_1 to Q_2 is sometimes called an *expansion in supply*.





Other Factors affecting the supply curve.

These are the factors that lead to shift the supply curve. Like demand, supply is not simply determined by price. The other determinants of supply are as follows:

- The costs of production (C). The main reasons for the change in costs are as follows:
 - a) Change in input prices: costs of production will rise if wages, rents, interest rates or any other input prices rise.
 - b) Change in technology can fundamentally alter costs of production
 - c) Government policy: costs will be lowered by government subsidies and raised by various taxes
- The profitability of alternative products (substitutes in supply) (P). If a product which is a substitute in supply becomes more profitable to supply than before, producers are likely to switch from the first good to this alternative. Other goods are likely to be more profitable if their prices rise or their costs of production fall.
- Nature random shocks and other unpredictable events. In this category we would include the weather and diseases affecting farm output, wars affecting the supply of imported raw materials the breakdown of machinery, industrial disputes, earthquakes, floods, fire etc (R).
- The aims of producers. A profit-maximizing firm will supply a different quantity from a firm that has a different aim, such as maximizing sales. For most of the time we shall assume that firms are profit maximisers (A).
- Expectation of future price changes. If prices are expected to rise, producers may temporarily reduce the amount they sell so that they would sell more in future after the price has gone up (E).
- The number of suppliers. If new firms enter the industry (market) supply is likely to increase and vice versa (NS).

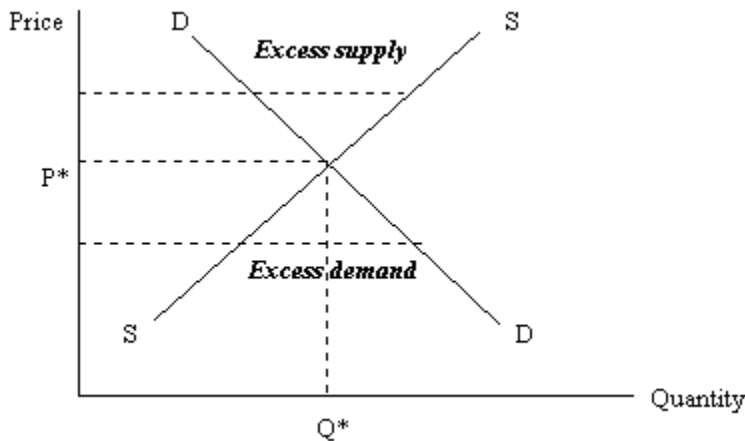
$$Q^S = f(C, P, R, A, E, NS)$$

MARKET EQUILIBRIUM

Market Price

At prices above the equilibrium (P^*) there is *excess supply* while at prices below the equilibrium (P^*) there is *excess demand*. The effect of excess supply is to force the price down, while excess demand

creates shortages and forces the price up. The price where the amount consumers want to buy equals the amount producers are prepared to sell is the *equilibrium market price*. All these situations are shown in the diagram below:

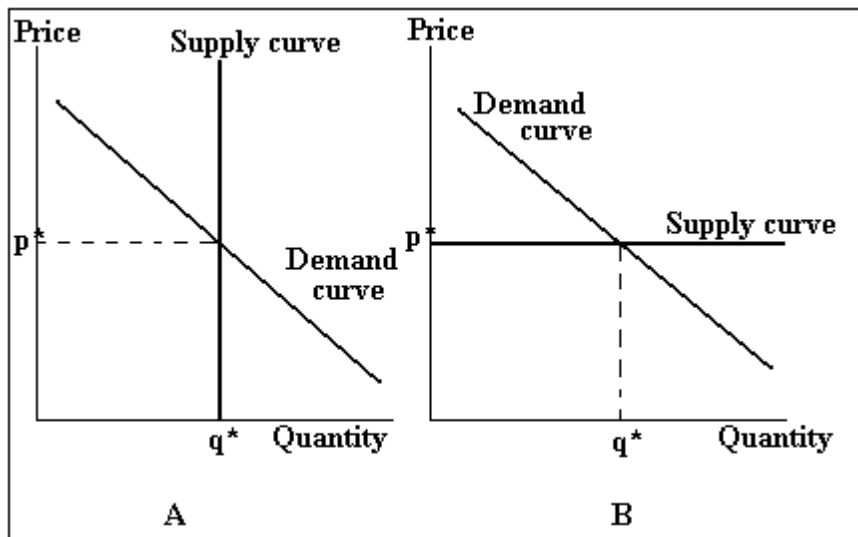


TWO SPECIAL CASES OF MARKET EQUILIBRIUM.

There are two special cases of market equilibrium that are worth mentioning since they come up fairly often. The first is the case of fixed supply. Here the amount supplied is some given number and is independent of price; that is, the supply curve is vertical. In this case the equilibrium quantity is determined entirely by the supply conditions and the equilibrium *price* is determined entirely by demand conditions.

The opposite case is the case where the supply curve is completely horizontal. If an industry has a perfectly horizontal supply curve, it means that the industry will supply any amount of a good at a constant price. In this situation the equilibrium price is determined by the supply conditions, while the equilibrium quantity is determined by the demand curve.

The two cases are depicted below. In these two special cases the determination of price and quantity can be separated, but in the general case the equilibrium quantity and price are jointly determined by the demand and supply curves.

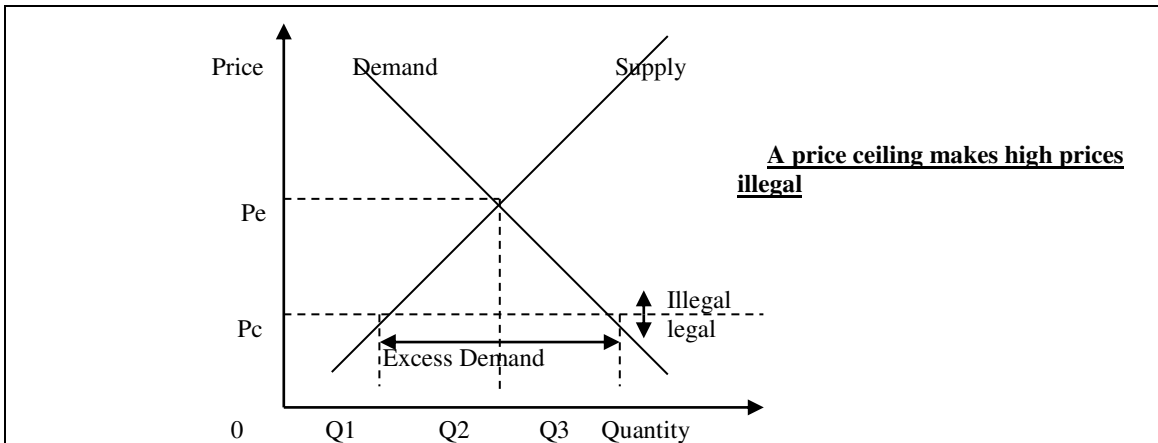


Special cases of equilibrium. Case A shows a vertical supply curve where the equilibrium price is determined solely by the demand curve. Case B depicts a horizontal supply curve where the equilibrium price is determined solely by the supply curve.

EFFECTS OF GOVERNMENT INTERVENTION: PRICE CONTROLS.

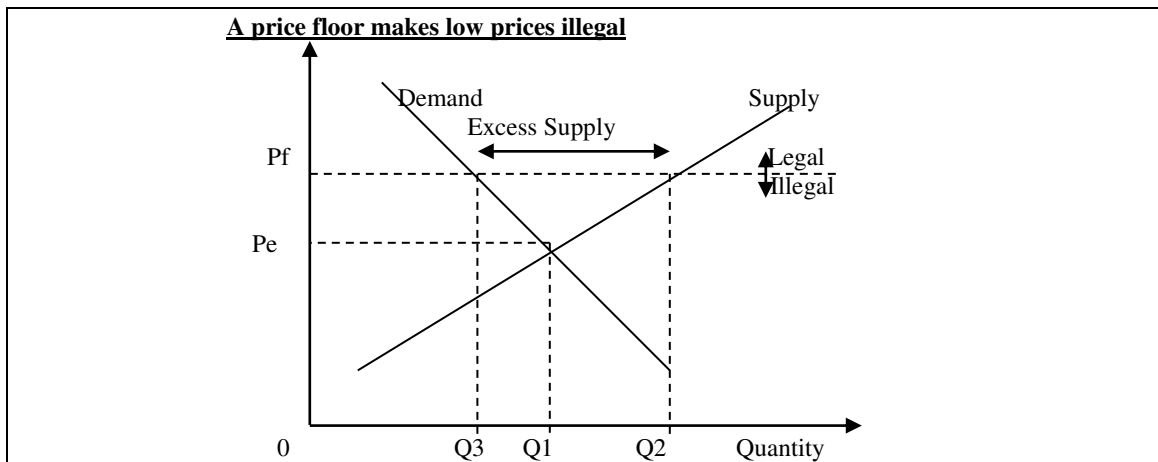
There have been many cases throughout history in which governments have been unwilling to let markets adjust to market-clearing prices. Instead, they have established either **price ceilings**, which are prices above which it is illegal to buy or sell, or **price floors**, which are prices below which it is illegal to buy or sell.

If a price ceiling is placed below the market-clearing price, as **P_c** is in the picture below, the market-clearing price of **P_e** becomes illegal. At the ceiling price, buyers want to buy more than sellers will make available. In the graph, buyers would like to buy amount **Q₃** at price **P_c**, but sellers will sell only **Q₁**. Because they cannot buy as much as they would like at the legal price, buyers will be out of equilibrium. The normal adjustment that this disequilibrium would set into motion in a free market, an increase in price, is illegal; and buyers or sellers or both will be penalized if transactions take place above **P_c**. Buyers are faced with the problem that they want to buy more than is available. This is a **rationing** problem.



Price ceilings are not the only form of price controls governments have imposed. There have also been many laws that establish **minimum** prices, or **price floors**. The graph below illustrates a price floor with price **P_f**. At this price, buyers are in equilibrium, but sellers are not. They would like to sell quantity **Q₂**, but buyers are only willing to take **Q₃**. To prevent the adjustment process from causing price to fall, government may buy the surplus, as the Zimbabwean government has done in agriculture and in precious metals. If it does not buy the surplus, government must penalize either buyers or sellers or both who transact below the price floor, or else price will fall. Because there is no one else to absorb the surplus, sellers will.

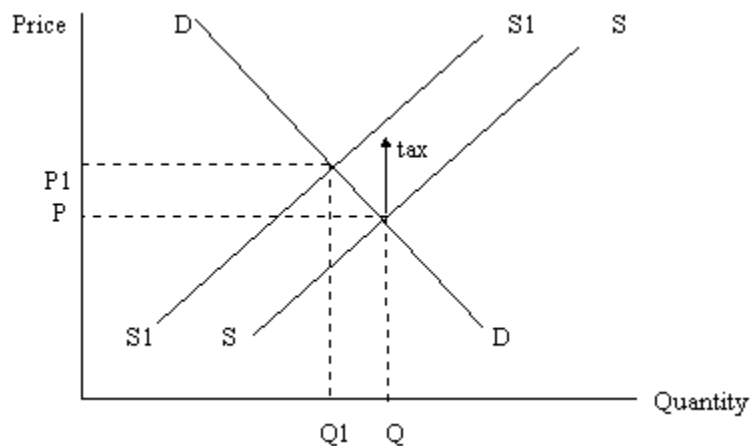
Rationing is necessary to deal with scarcity. When an item is scarce, people must sacrifice something in order to get as much of the item as they would like to have. There are some goods that are not scarce. Air is an example--it is free to all who want to breathe it. Ice is not scarce in Greenland. But almost all other goods are scarce. Price is a way to ration goods. It deprives those who do not have enough income or desire for a product. The function of price as a rationer is most clearly seen when price is prohibited from acting as a rationer, so that some other method of rationing is used.



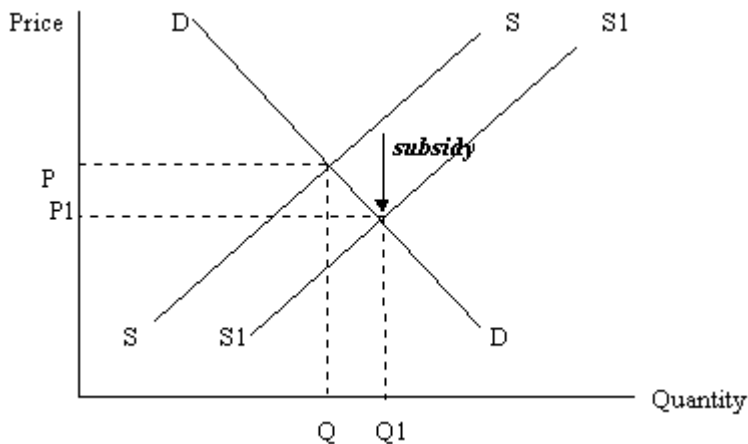
Price floors in Zimbabwe have been observed in the labour market where the government sets the minimum wages. The graphical analysis is similar to the one above, but with wage on the price axis and quantity of labour on the quantity axis.

Indirect Taxes and Subsidies

In the figure below an indirect tax has been added to SS. This has the effect of shifting the supply curve up vertically by the amount of the tax. Note in the diagram below that price does not increase by the full amount of the tax. This suggests that the firm pays part of the tax.



In this figure a subsidy has been given to the firm. This has the effect of making firms willing to supply more at each price and so shifts the supply curve downwards. The shift is equivalent to the value of the subsidy. Note that price falls by less than the full amount of the subsidy. This suggests that the firm keeps part of the subsidy.



ELASTICITY OF DEMAND

Elasticity is a measure of sensitivity of one variable to changes in another variable. It is also defined as the degree of responsiveness of one variable to changes in another variable.

Price elasticity of demand (PED) - measures the sensitivity of quantity demanded to price changes. It tells us what percentage change in the quantity demanded for a good will be following a one percent increase/decrease in the price of that good.

Let's denote quantity and price by Q and P , to give us the expression for price elasticity of demand.

$$PED = \frac{\Delta Q}{Q} \div \frac{\Delta P}{P} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

Example:

Suppose that at a price of \$100 monthly sales of bicycles in a city are 1000. Next month the price of a bicycle goes up to \$101. As a result of a price increase the quantity of bicycles demanded per month falls to 990.

The percentage change in price is therefore $\frac{\Delta P}{P} \times 100\% = \frac{101 - 100}{100} \times 100\% = 1\%$

The percentage change in quantity demanded is

$$\frac{\Delta Q}{Q} \times 100\% = \frac{990 - 1000}{1000} \times 100\% = -0.5\%$$

The price elasticity of demand is usually a negative number. When the price of a good increases the quantity demanded usually falls.

$$PED = 0.5\% \div 1\% = 0.5$$

Interpretation of own-price elasticity of demand

If $PED > 1$ then demand is elastic, typically for luxury goods.

If $PED < 1$ then demand is inelastic, typically for basic commodities and habit-forming goods.

Determination of Price Elasticity of Demand.

- *The availability of substitute goods*

The more substitute goods there are for a good, whose price rises, the more elastic is the demand for good.

- *The period of adjustment to price changes.*

The demand for a good is generally more elastic in the long run than in a short run because people generally find more substitutes for a good as time goes by.

- *The portion of consumer budget allocated to the product.*

Large percentage increases in the price of goods that constitute small portions of your total budget might have little effect on your purchases of these goods if you regard them as necessities.

Income elasticity of demand (YED).

Income elasticity of demand is the degree of responsiveness of quantity demanded to changes in income.

$$YED = \frac{\Delta Q}{Q} \div \frac{\Delta Y}{Y} = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q}$$

If $YED > 0$ normal good

If $YED < 0$ inferior good

If $YED = 0$ income elasticity for goods whose consumption is completely unresponsive to changes in income e.g. necessities like salt.

If $YED > 1$ luxury goods, because as income increases the share of those goods also increases. (e.g. foreign travel)

Example.

Suppose the consumer is consuming apples and oranges. Price of an apple is \$5 and the price of an orange is \$40. The demand for apples is 56 units while the demand for oranges is 87 units. The consumer has an income of \$200. Suppose the consumer's income increased to \$300 while the demand for oranges has decreased to 70 units and the price of apple has decreased to \$2.

(a) Calculate the income elasticity of demand for oranges.

$$YED = \frac{\Delta Q}{Q} \div \frac{\Delta Y}{Y} = \frac{70 - 87}{87} \div \frac{300 - 200}{200} = \frac{-0.2}{0.5} = -0.4$$

Cross elasticity of demand (CED).

CED is the degree to which quantity demanded of one good responds to changes in the price of another good. It may be therefore expressed as follows:

$$CED = \frac{\% \Delta Q_x}{\% \Delta P_y}$$

$$CED = \frac{\Delta Q_X}{Q_X} \div \frac{\Delta P_Y}{P_Y} = \frac{\Delta Q_X}{\Delta P_Y} \times \frac{P_Y}{Q_X}$$

Complementary and substitute goods

There are two values of CED to consider:

- **CED positive:**

This is the case for substitute goods. Substitute goods are goods that are used for exactly the same purpose. An increase in the price of one good leads to an increase in quantity demanded of its substitute.

Illustration

	Old price and quantity		New price and quantity	
	\$	kg	\$	kg
Butter	1.90	4000	2.1	3000
Margarine	1.30	6000	1.30	7000

Solution

$$CED = \frac{7000 - 6000}{6000} \div \frac{2.10 - 1.90}{1.90}$$

$$CED = 1.58$$

- **CED negative:**

This is the case for complementary goods, which are consumed together. An increase in the price of one causes a decrease in the quantity demanded of the other.

Illustration

	Old price and quantity		New price and quantity	
	\$		\$	
Bread	60	600 000	90	400 000
Butter	50	100 000kg	50	67 000 kg

Solution

$$CED = \frac{67000 - 100000}{100000} \div \frac{90 - 60}{60}$$

$$CED = -0.66$$

Significance of elasticity

Business

If the demand curve is inelastic then a decrease in price will lead to a fall in revenue and vice versa. A decrease in price will increase revenue if the product's demand is elastic and an increase in price will reduce revenue.

E.g. Government will always wish to tax inelastic goods e.g. cigarettes and alcohol because a tax on this type of goods does not reduce demand very much.

ARC ELASTICITY OF DEMAND

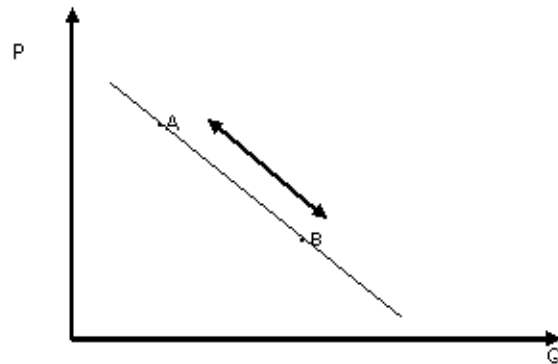
Arc price elasticity of demand measures the elasticity of demand over an interval on the demand function. Instead of using the price and quantity at a point as in point elasticity, arc elasticity uses the average of the prices and quantities at the beginning and the end of the stated interval.

$$AED = \frac{\Delta Q}{\Delta P} \times \frac{\frac{1}{2}[P_1 + P_2]}{\frac{1}{2}[Q_1 + Q_2]} = \frac{\Delta Q}{\Delta P} \times \frac{P_1 + P_2}{Q_1 + Q_2}$$

This formula is often referred to as the midpoints elasticity formula or arc price elasticity formula.

NB. For linear functions (e.g. $Q = a - bP$), the arc- price elasticity of demand formula can also be written as:

$$AED = \frac{\Delta Q}{\Delta P} \times \frac{P_1 + P_2}{Q_1 + Q_2} = -\frac{1}{b} \times \frac{P_1 + P_2}{Q_1 + Q_2} \quad \text{since} \quad \frac{\Delta Q}{\Delta P} = -\frac{1}{b}$$



We can calculate arc elasticity between points A and B.

CHAPTER 3: CONSUMER BEHAVIOUR

CONSUMER AND MARKET BEHAVIOR

The concept of utility can help us understand two related aspects of consumer behavior.

- It enables us to predict how an individual will allocate his expenditure, given a fixed income between goods and services available for consumption.
- It enables us to predict the effect of a price change on the quantity demanded of a good and so confirms the law of demand.

TOTAL AND MARGINAL UTILITY

UTILITY is the term used by economists to convey the pleasure and satisfaction derived from the consumption of goods and services. Utility represents the fulfillment of a need or desire through the activity of consumption.

We must assume that it is possible to quantify and measure changes in satisfaction or utility. For this purpose a “util” will be used as a measure for utility. In reality, utility is a psychological concept and its subjective nature makes it unmeasurable. Nevertheless, we shall ignore this and proceed as if utility can be measured in utils just like distance can be measured in meters or temperature in degrees. The standard util is totally imaginary.

TOTAL UTILITY represents the satisfaction gained by a consumer as a result of his overall consumption of goods.

MARGINAL UTILITY represents the change in satisfaction resulting from the consumption of a further unit of a good.

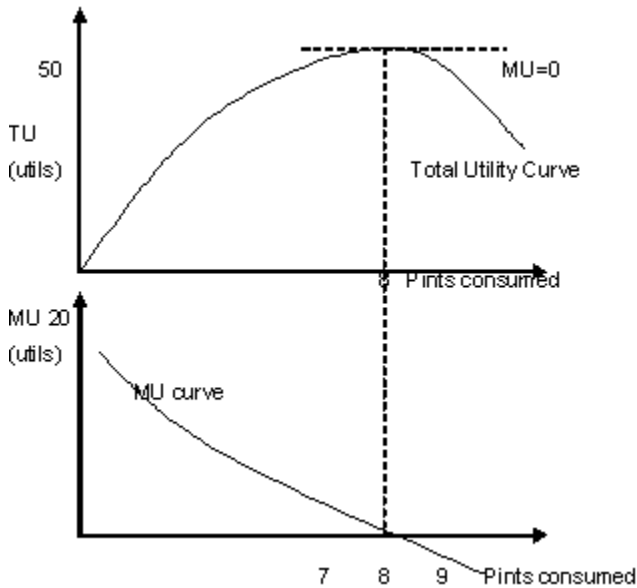
Assuming that utility can be measured, we can say, for instance, that a given individual enjoys 37 units of satisfaction (utils) from drinking 3 pints of beer during an evening. This is a measure of total utility. If one more pint increases his total utility to 42 utils the marginal utility of his fourth pint would be equal to 5 units of satisfaction. The marginal utility of the fourth pint equals the total utility derived from 4 pints minus total utility derived from 3 pints. Marginal utility is the increase in total utility that results from the consumption of one more unit.

An individual's utility schedule

Pints drunk in an evening	Total utility	Marginal Utility
0	0	-----
1	17	17
2	29	12
3	37	8
4	42	5
5	45	3
6	47	2
7	48	1
8	48	0
9	47	-1

The figures clearly display a crucial element of utility theory: the law of Diminishing Marginal Utility. The law states that the satisfaction derived from the consumption of an additional unit of a good will decrease as more of the good is consumed, assuming that the consumption of all other goods is held constant. Table above satisfies this law in that although each pint consumed until the ninth pint adds to total satisfaction, it does so by decreasing amounts. While the third pint adds 8 units of satisfaction,

the 4th pint only adds 5 units. Neither of these can compare with the first pint that resulted in 17 units of satisfaction. It's also interesting to note that MU can be negative. If the individual were forced to drink the ninth pint, his total utility would actually be reduced. This is sometimes called disutility.



It is important to appreciate fully the implications of the distinction between total and marginal utility. If you were given the choice of giving up totally your consumption of either water or petrol, you would choose to give up petrol. The implication is that water provides you with more total utility than petrol.

A man dying of thirst in the desert is faced with different conditions and therefore different marginal utilities. He would definitely place more value on an extra gallon of water. From these examples we can see that when a consumer makes a decision, he is concerned with the relative utilities of different goods. But given the availability of resources, economic behavior will be determined by relative marginal utilities rather than total utilities: shall I consume a few extra units of good A at the expense of good B.

THE INDIFFERENCE CURVE

An indifference curve represents all combinations of market baskets that provide the same level of satisfaction or utility to a person or consumer. Assume that you have a basket of goods containing 16 eggs and 6 bags of crisps. If someone comes along with a basket containing 20 eggs and 5 bags of crisps and offer to change baskets, would you accept, refuse or find it impossible to decide, as you would be indifferent between the two baskets?

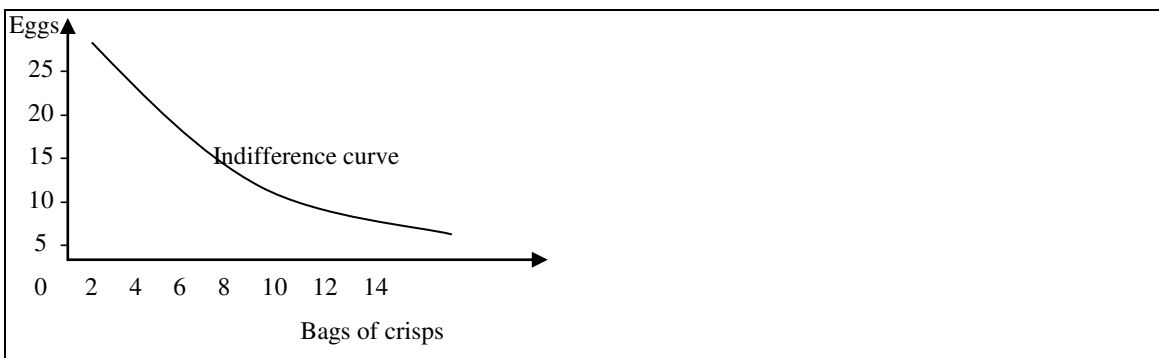
Alternatively, what would be the minimum no of eggs you would require in order to compensate exactly for the loss of your 6 bags of crisps? If your answer to this second question is 3 eggs, you are in effect saying that you are indifferent between a basket containing 16 eggs and 6 bags of crisps and one containing 19 eggs and 5 bags of crisps: each would give you the same total utility. This being the case, the original offer would certainly have been accepted as 20 eggs and 5 bags of crisps clearly represent a more attractive combination.

In this way it is possible to build up a set of combination of any two goods between which the consumer is indifferent. Each combination will provide the same total utility. This whole operation can be carried out without ever having to put a precise figure on the amount of utility involved.

AN INDIFFERENCE SCHEDULE

<u>Basket</u>	<u>Eggs</u>	<u>Bags of crisps</u>
A	22	4
B	19	5
C	16	6
D	13	7
E	10	8
F	7	9

AN INDIFFERENCE CURVE



This curve is called the indifference curve and each point on the curve represents a combination of eggs and bags of crisps between which this particular individual is indifferent. The nature of the curve reveals various aspects of consumer behavior. Consider the following indifference curve.

Some basic assumptions.

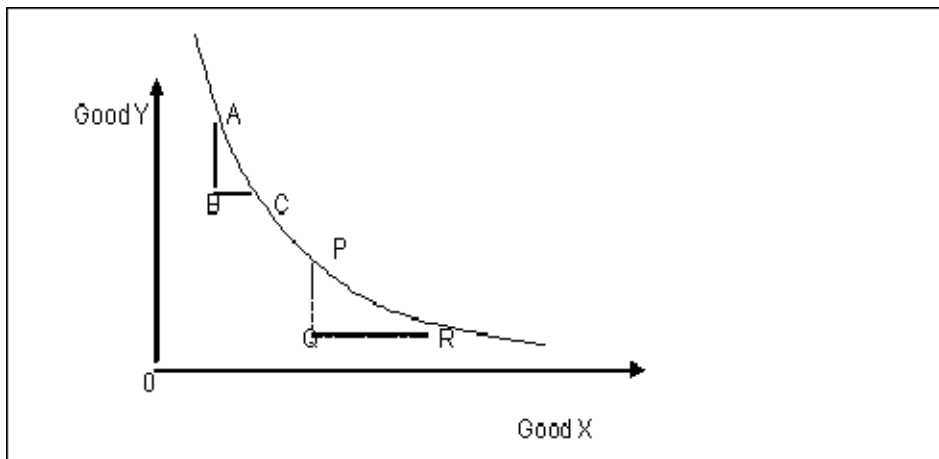
- *Preferences are complete*, which means that a consumer can rank all market baskets. E.g. For any two market baskets A and B, the consumer can prefer A to B or B to A, or can be indifferent.
- *Preferences are transitive*. It means, that if a consumer prefers basket A to B and prefers B to C, then he should also prefer A to C.
- *Preferences are reflexive (desirable)*. So that leaving costs aside, consumer always prefers more of any good to less. (this applies to economical goods, and not applies to bad goods such as pollution)

Marginal Rate of Substitution

$$AB = PQ$$

$$BC < QR.$$

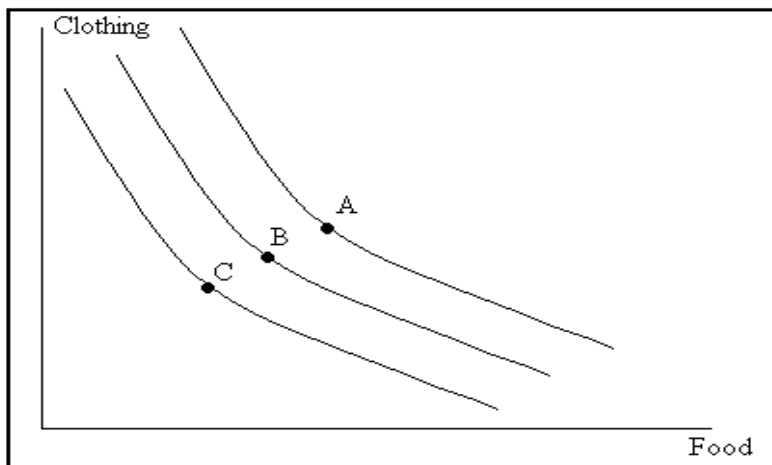
The slope of the curve between A and C is given by the ratio AB/BC represents the amount of good X necessary to make up for the loss of AB of good Y that would result if the individual moved from combination A to combination C. The rate at which one good can be substituted for another without any change in total utility is called the Marginal Rate of Substitution (MRS). The MRS between P & R is less than the MRS between A & C. The fact that this ratio decreases as one moves down the curve from left to right is known as the DIMINISHING MARGINAL RATE OF SUBSTITUTION.



Another point to notice is that an IC slopes downwards from left to right. Assuming that both goods are desirable, the rational consumer could not be indifferent to a basket containing more of both goods. Therefore, as we move from one point to the other on the IC, while the quantity of one good increase, the quantity of the goods has to decrease if total utility is to remain unchanged. This is why the slope of the IC is normally negative.

THE INDIFFERENCE MAP

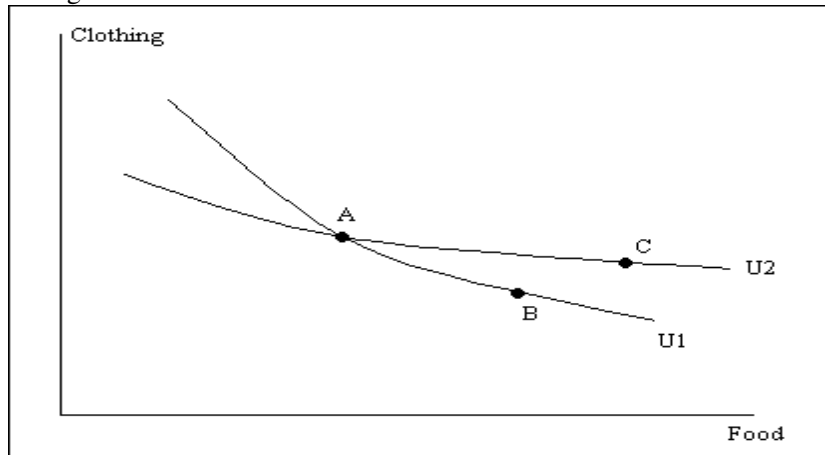
It is a set of indifference curves that describes the person's preferences. A set of indifference curves represents an ordinal ranking. An ordinal ranking arrays market baskets in a certain order, such as most preferred, second most preferred... 3d most preferred.



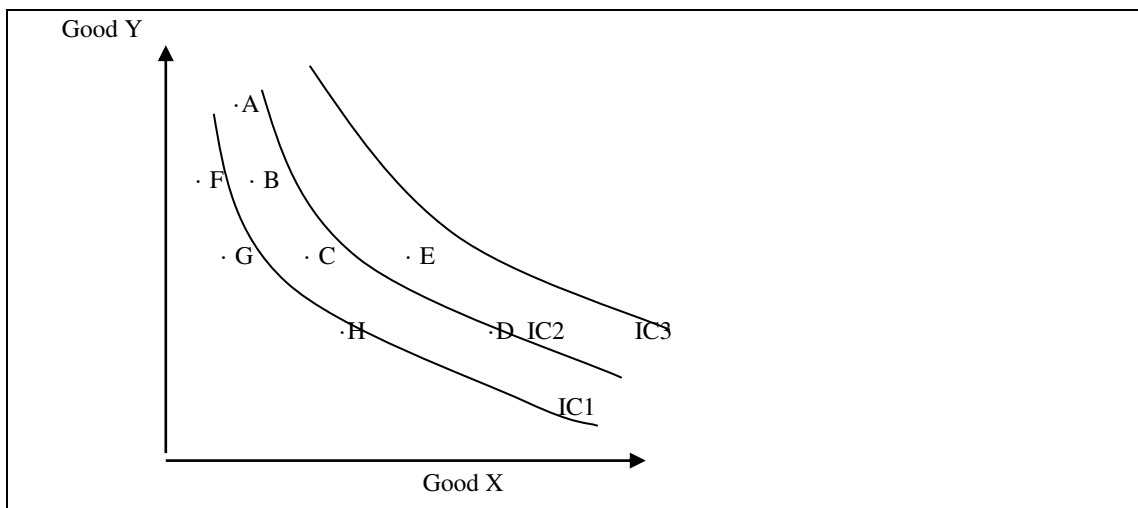
Market basket A is in the highest of three indifference curves. It is preferred to B and C, B is preferred to C.

The assumptions made about the consumer preferences for economic goods imply that indifference maps have the followings:

- Market baskets on indifference curves further from the origin are preferred.
- Marginal rate of substitution (MRS). MRS_{XY} is rate of substitution of X for Y. It is the amount of good Y that the consumer would give up to obtain one more unit of good X while holding utility constant. The slope of an indifference curve measures the consumer MRS between two goods.



A single IC represents but one possible level of total utility. In fig below A, B, C & D all represent identical levels of total utility (IC). If point E were taken at random, then together with all other points which provide an identical utility it would form a second IC (IC2). Clearly, all the combinations given by points on IC2 would provide a higher level of total utility than given by IC1.



Point F would represent a lower level of total utility than any point on IC1 and IC2, but an equal level of utility when compared to any other point on IC3 e.g. G and H.

There is an infinity number of ICs; each represents a different level of total utility. A representative sample of a consumer's many ICs over a given time period is called an indifference map.

THE BUDGET LINE

- Account for the fact that the consumers face budget constraints

$$P_X X + P_Y Y = B$$

Where P_X and P_Y are the prices for goods X and Y respectively.

X and Y are goods X and Y respectively, and

B is the consumer's budget.

They have limited income to allocate among consumption items.

While an IC describes a consumer's preferences, **a budget line** shows the various combinations of the two goods that can be bought at current prices with a fixed budget or income. Assume an individual has 3 dollars a week to spend on oranges and bags of crisps, and that eggs cost 10c each while crisps cost 15c a bag. If the individual spends his entire budget on eggs he can afford 30. If he can spend it all on crisps, he can afford 20 bags.

Between these two extremes there is a variety of other possibilities e.g. 15 eggs + 10 bags of crisps. Each point on the budget line represents one of the several possible combinations that will cost exactly 3 dollars.

If X represents the number of bags of crisps and Y the number of eggs consumed per time period, we can write the equation of the budget line as:

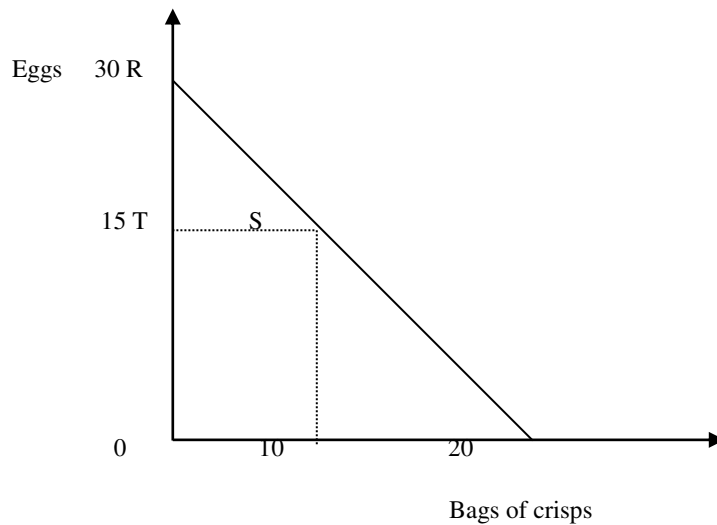
- $15X + 10Y = 300$

X bags of crisps at 15c each and Y eggs at 10c each must come to a total of 300c or

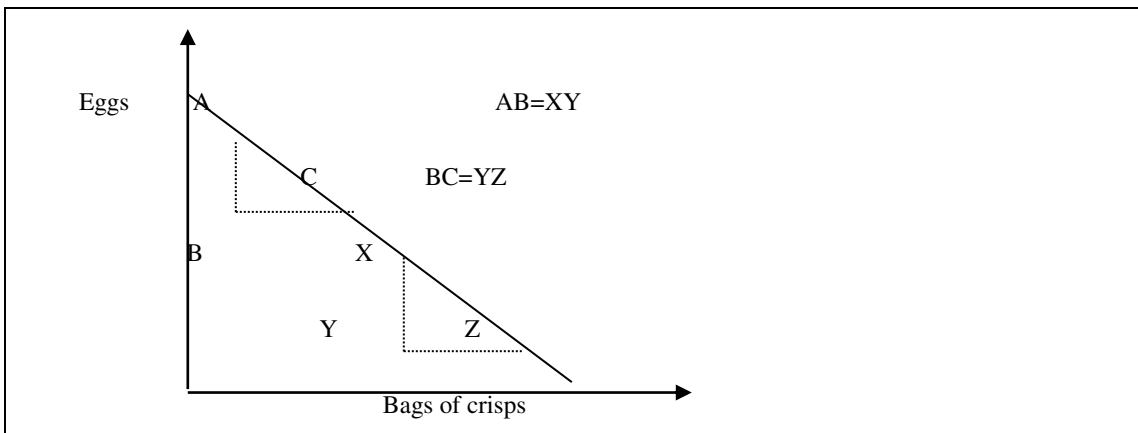
3 dollars.

In the light of market prices, the slope of the budget line is the amount of one good that has to be sacrificed in order to buy an additional unit of the other good. This will be the same for any point on the budget line. If this consumer reduces his consumption of eggs by AB he will save enough to buy BC bags of crisps. While the IC slope tells us of the rate at which the consumer is willing to trade one good for the other, given his preferences the budget line tells us the current market rates of exchange given existing prices.

A BUDGET LINE



THE SLOPE OF THE BUDGET LINE



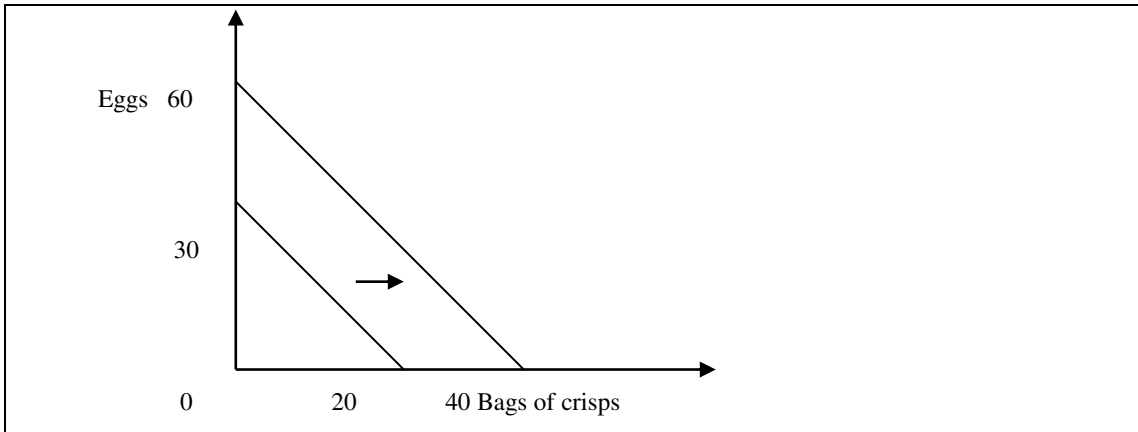
The slope of the budget line = $AB/BC = XY/YZ$.

Shifts in the budget line

The budget line will shift its position as a result of any change in the consumer's budget or changes in the prices of either of the two goods under consideration.

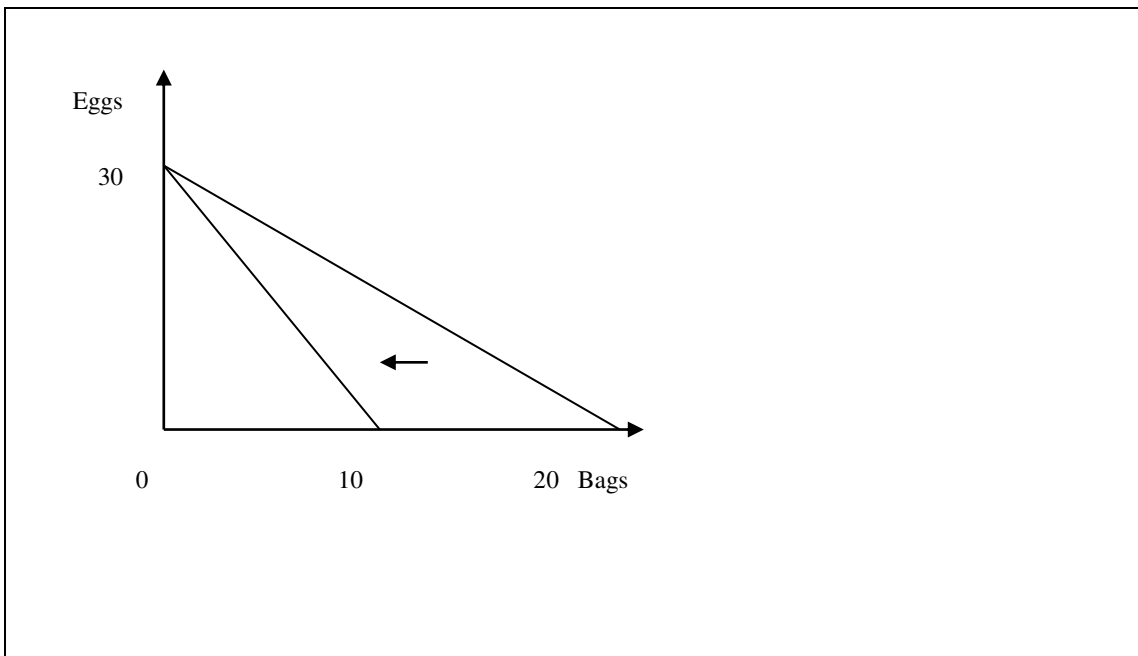
If the consumer's budget is doubled, he could now buy 60 eggs if no of crisps are consumed or 40 bags of crisps if no of eggs are consumed. An increase in income will shift the budget line away from the origin, while a fall in income will shift towards the origin. Unless the **relative prices change** the new budget lines will be parallel to the original one.

a) An increase in income



If income and the price of eggs remain constant while the price of crisps doubles, the new budget line will be as shown below.

b) An increase in the price of crisps



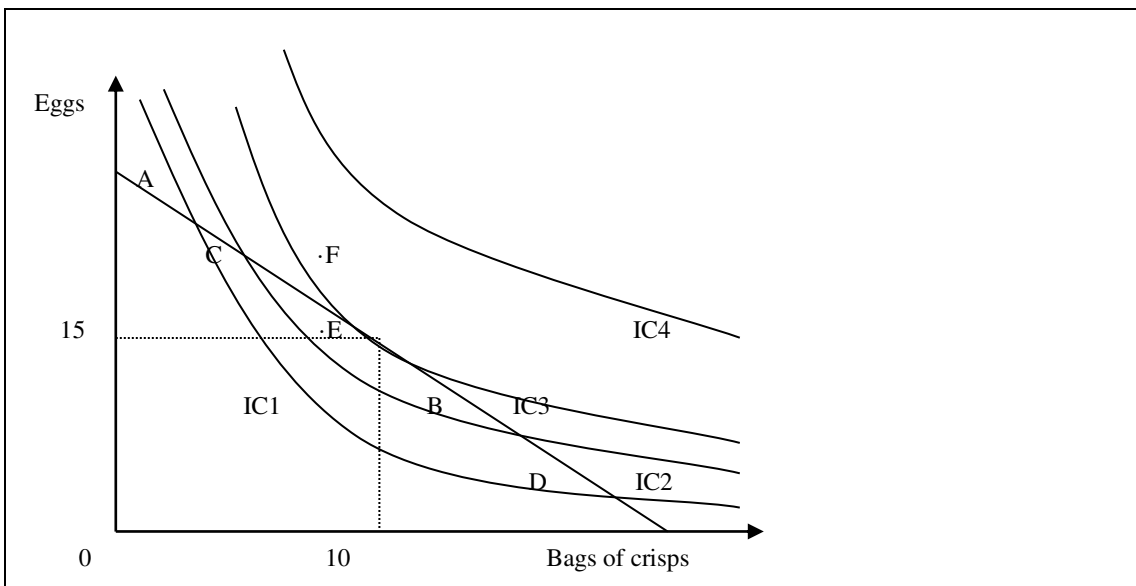
Changes in relative prices will alter the slopes of the budget lines.

CONSUMER EQUILIBRIUM

Having explained both indifference curves and budget lines, we are now in a position to represent consumer equilibrium graphically. By drawing an individual's budget line and indifference map on the same graph, consumption possibilities and preference can be compared.

Given the constraint of his budget line the individual's aim is to maximize his total utility. Each point on the budget line represents a combination of eggs and bags of crisps that he can afford. He is looking for the point that lies on the IC that is as far as possible from the origin. The further the IC is from origin, the higher is the level of total utility it represents. In this way **point B** is preferable to **point A**, as it lies on IC2, which is further from the origin IC1. The consumer will be indifferent between point B and C as they both lie on IC2. Out of all the points on the budget line point E will bring the greatest satisfaction as all other points on the budget line lie on ICs which are nearer to the origin.

CONSUMER EQUILIBRIUM



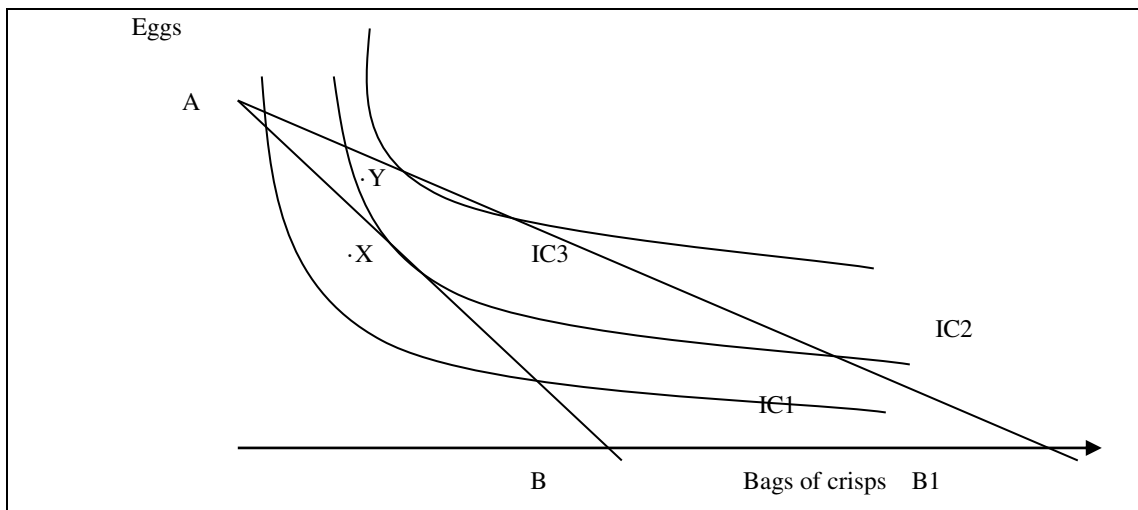
In the diagram above the consumer is in equilibrium i.e. (maximising his total utility, given his fixed budget) when he is consuming 15 eggs & 10 bags of crisps per time period. This is given by point E on the budget line, the point where the budget line is just tangential to one of the IC.

Point F is clearly preferable to point E, but given the current market prices and a fixed budget, the combination lies outside the consumer's range of consumption possibilities. CONSUMER EQUILIBRIUM is represented graphically by the point of tangency of the budget line with an IC. At such a point of tangency, the slope of the budget line is equal to the slope of the IC. It follows that consumer equilibrium is reached when the ratio of the prices of the two goods is equal to the consumer's MRS ($P_X / P_Y = MRS_{XY}$).

INCOME AND SUBSTITUTION EFFECTS

- What happens if the price of bags of crisps falls from \$2 to \$1?
- The budget line shifts from AB to AB1. As a result of that change, the optimal combination of eggs and bags of crisps shifts from X to Y.

The impact of a price change



In the example above more bags of crisps are bought when their price falls and more eggs are also bought.

Two things happen when the price of a good falls. First the fall of price has led to an increase in the real income of the consumer and this change in income will alter the goods that the consumer chooses to purchase. This is the **INCOME EFFECT**. Second, the relative prices of eggs and bags of crisps have changed, which also alter the choices of purchases. This is the **SUBSTITUTION EFFECT**.

In the diagram below the overall change induced by the fall in the price of bags of crisps (the price effect) is indicated by the shift from X to Y.

- IC2 represents higher real income than IC1 before the price fall.
- The slope of the budget line represents relative prices.

$(X-Y)$ = Price effect

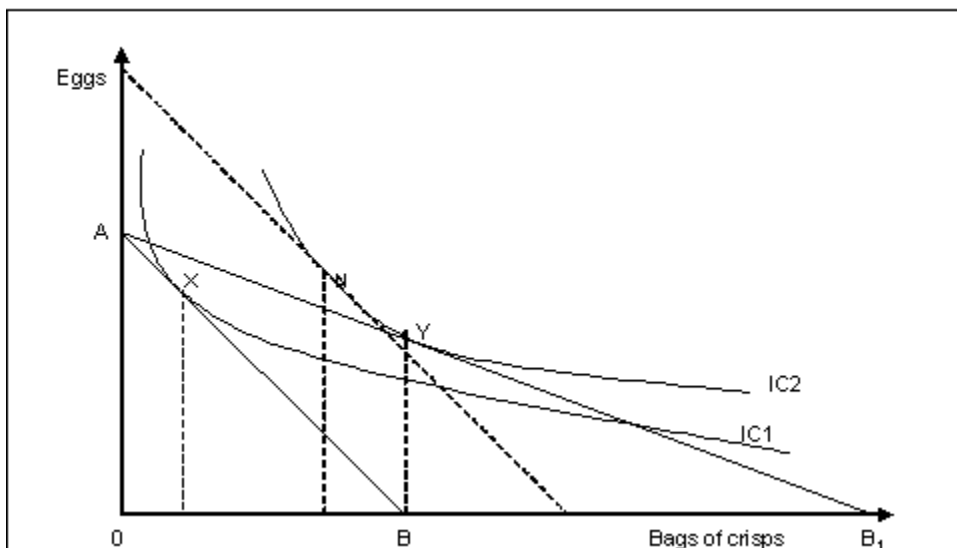
$(X-N)$ = Income effect

$(N-Y)$ = Substitution effect

$(X-Y) = (X-N) + (N-Y)$

Total Effect = Income Effect + Substitution Effect.

THE INCOME & SUBSTITUTION EFFECTS



The substitution effect may be defined as the change in the basket of goods that would be purchased if relative prices change at a constant real income. This may be interpreted as the movement around the indifference curve, IC2 from N to Y. The above analysis is for a normal good.

As for the inferior goods less of them are demanded as real income increases. Note that we have another type of inferior good called a Giffen good. For this type of good the negative income effect offsets the positive substitution effect to the extent that less than the initial quantity of the good is consumed.

CHAPTER 4: PRODUCTION

Production is the process of using the services of labor and equipment together with national resources and materials to make goods / services available.

Technology is the knowledge of how to produce goods and services.

Production function is the relationship between inputs and the maximum attainable output under a given technology.

In order to understand the economics of production, we have to start by examining the purely physical aspects; i.e. the relationship between the units of capital, land, and labor employed and the resultant physical units of output. In making a product, a firm does not have to combine the inputs in fixed proportions. Many farm crops can be grown by using relatively little labor and relatively large amounts of capital (machinery, fertilizers etc) or by combining relatively large amounts of labor with very little capital. In most cases the firm has some opportunity to vary the input mix.

The effects of varying the proportions between the factors of production is a subject of great importance because nearly all short run changes in production involve some changes in these

proportions. When a firm wishes to increase (decrease) its output it cannot, in the short run, change its fixed factors of production, but it can produce more (less) by changing the amounts of the variable factors (labor, materials etc). When the farmers wish to increase their output they are usually obliged to do so by using more labor, more seed, more fertilizer (i.e. the variable factors) on some fixed supply of land (the fixed factor).

Manufacturers are in a similar position. In the short run they cannot extend their factories or install more machinery but they can adjust their output by varying the quantities of labor raw materials fuel and power.

The short run is a period of production during which some inputs cannot be varied. In the short run for example manufacturing firms are confined to a given size of factory. □

The long run is a period of production so long that producers have adequate time to vary all their inputs used to produce a certain commodity. □

Total product of a variable input is the amount of output produced where a given amount of that input is used along with the fixed inputs.

The average product of variable input the total product of the variable input divided by the amount of that input used.

$$AP_L = \frac{TP_L}{L}$$

Marginal product of variable input is the change of the TP corresponding to one unit change in the input.

$$MP_L = \frac{\Delta TP_L}{\Delta L}$$

NON-PROPORTIONAL RETURNS

Labor hours L	Total product of labor TP ₁	Average product of labor AP ₁ =TP ₁ /L	Marginal product MP ₁ = $\frac{\Delta TP_1}{\Delta L}$
1	10	10	
2	26	13	16
3	56	18.6	30
4	84	21	28
5	97	19.4	13
6	102	17	5
7	102	14.6	0
8	98	12.25	4

Table above illustrates some important relationships, but before we examine them we must state the assumptions on which the table is based.

- Labor is the only variable factor.
- All units of the variable factor are equally efficient.
- There are no changes in the techniques of production.

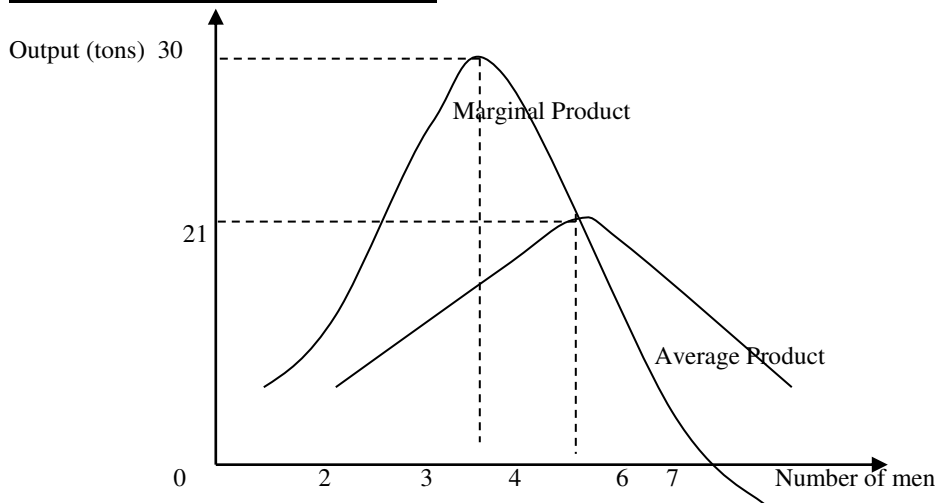
On the basis of these assumptions we can conclude that any changes in productivity arising from variations in the number of people employed are due entirely to the changes in the proportions in which labor is combined with other factors. Table above illustrates the Law of Diminishing Returns (or The Law of Variable Proportions) which states that “As we add successive units of one factor to fixed amounts of other factors the increments in total output will at first rise and then decline.”

Returns to the variable factor

Since labor is the only variable factor, changes in output are related directly to changes in employment so that we speak of changes in productivity of labor or changes in the **returns to labor**. As the number of people increases from 1 to 6, total output continues to increase, but this is not true of the average product (AP) and the marginal product (MP). As more people are employed, both the AP and the MP begin to rise, reach a maximum and begin to fall. In figure below as the number of people increases from 1 to 3 the marginal product of labor is increasing. Up to this point the fixed factors are being underused-the people are too thin on the ground.

When the number of people employed exceeds 3 the marginal product of labor begins to fall an indication that the proportions between the fixed and variable factors are becoming less favourable. Marginal product begins to fall before average product and we get the maximum average product of labor when 4 people are employed. If we now wished to increase output and maintain the same level of productivity of labor it is obvious that an increase in the fixed factors must accompany the increase in the variable factors. This would be a change of scale and is the subject of the next section.

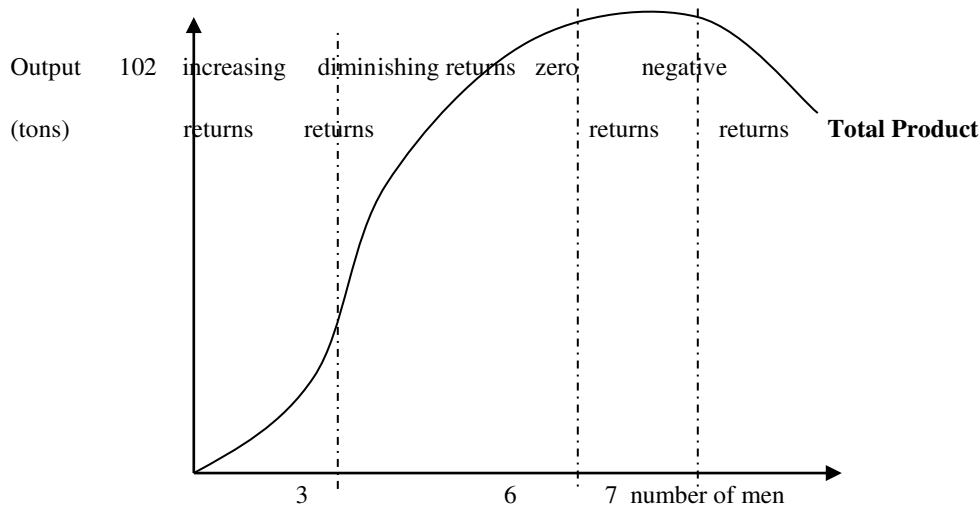
Average and Marginal Productivity



NB The MPs are plotted at the midpoints because they refer to the change in TP as employment changes.

It is this feature of increasing production and falling productivity that is highlighted by the Law of diminishing Returns. In table of figures above we see that Diminishing Marginal Returns set in after the employment of the third person and Diminishing Average Returns after the employment of the fourth person. Note that the marginal productivity of the seventh person is zero-his employment does not change total output. This may not be so unrealistic as it appears. In some underdeveloped lands where peasant families are confined to their individual plots, it's quite conceivable that the marginal productivity of very large families is zero.

Figure below makes use of the total product curve and provides another view of the relationships between employment and output where some of the factors are fixed in supply.



We can summarise the possible effects of increasing the quantity of variable factors as follows:

- Increasing Returns - Total product increases at an increasing rate (MP is increasing).
- Constant Returns (not illustrated) – total product is increasing at a constant rate (MP is constant).
- Diminishing Returns – total product is increasing at a decreasing rate (MP is falling).
- Zero Returns – total product is constant (MP is zero).
- Negative Returns – total product is falling (MP is negative).

It is important to note that although the illustration used above have concentrated on labor as the variable factor, the law of variable proportions (or diminishing returns) is equally applicable to land and capital, and no doubt to entrepreneurship. The marginal and average productivity of capital will at some point, start to decline as more and more capital is applied to a fixed supply of land and labor. The same will apply to the productivity of land as more and more land is combined with a fixed amount of labor and capital.

The Law of diminishing returns only applies **when other things remain equal**. The efficiency of the other factors and the techniques of production are assumed to be constant. Now we know that these other things do not remain constant and improvements in technical knowledge have tended to offset the effects of the law of diminishing returns. Improved methods of production increase the productivity of the factors of production and move the AP and MP curves upwards. But this does not mean that the law no longer applies.

It's true that in the short period (when other things change very little) increments in the variable factors will at some point yield increments in output which are less than proportionate. In some less developed regions where there is little or no technical change and population is increasing we can, unfortunately, see the law of diminishing returns operating only too clearly.

RETURNS TO SCALE

The law of diminishing returns deals with what are essentially short run situations. It is assumed that some of the resources used in production are fixed in supply. In the long run, however, it is possible for a firm to vary the amounts of all the factors of production employed; more land can be acquired, more buildings erected and more machinery installed. What we are saying is that, in the long run it is possible for a firm to change the SCALE OF ACTIVITIES. A change of scale takes place when quantities of all the factors are changed by some percentage so that the proportions in which they are combined are not changed. It is a feature of production that when the scale of production is changed, output changes are not usually proportionate. When a firm doubles its size, output will tend to change by more than 100% or less than 100%.

$$Q = f(L, K).$$

Where Q is the maximum amount under current technology that could be produced with any given combination of labor services L, and capital services K.

The production function can also be represented by the Cobb-Douglas Function which is written as follows:

$$Q = AL^a K^b$$

The marginal product of labor from this function is $MP_L = \frac{dQ}{dL} = aAK^b L^{a-1}$

The slope of the $MP_L = \frac{d(MP_L)}{dL} = a(a-1)AK^b L^{a-2}$

For us to talk about returns to scale we have to multiply all our factors of production by a scale factor; and we are going to use scale factor k to do that.

Initial output: $Q = AK^a L^b$

New output: $Q^* = A(kK)^a (kL)^b = AK^a L^b (k^{a+b})$

Using this equation we can now talk about the returns to scale.

If $(a + b) > 1$, we have increasing returns to scale. This implies that if inputs are each multiplied by factor k output will increase by more than factor k.

If $(a + b) = 1$, we have constant returns to scale. This implies that if inputs are each multiplied by factor k output will increase by factor k.

If $(a + b) < 1$, we have decreasing returns to scale. This implies that if inputs are each multiplied by factor k output will increase by less than factor k.

Those features of increasing size that account for increasing returns to scale are generally described as Economies of Scale. The causes of falling efficiency as the size of the firm increases are described as Diseconomies of Scale. For example while the inputs of land, labor and capital may be increased proportionately; this may not be possible with regard to management ability. The entrepreneurial skills required to manage large enterprises are, it seems, limited in supply so that it is often difficult to match the increase in the supply of other factors with a corresponding increase in the supply of management ability.

COSTS OF PRODUCTION

Total Costs

A firm organizes the manufacture of a good or service. An industry is made up of all those firms producing the same commodity. The amount spent on producing a given amount of a good is called total cost, TC, and is found by adding together variable costs (VC) and fixed costs (FC).

Variable costs

Variable costs depend on how many goods are being made (output). If just one more unit is made then the total variable costs rise. Variable costs include the following:

- Weekly wages paid to the shop floor workers.
- The cost of buying raw materials and components
- The cost of electricity and gas.

Fixed Costs

Fixed costs are totally independent of output. Fixed costs have to be paid out even if the factory stops production. Fixed costs include the following:

- Monthly salaries paid to managers;
- Rent paid for the use of premises;
- Rates paid to the council;
- Any interest paid on loans;
- Insurance payments in the case of accidents;
- Money put aside to replace worn-out machines and vehicles sometime in the future (depreciation).

Average Cost

Average Cost (AC) or cost per unit is the cost of producing one item and is calculated by dividing total costs by total output.

Marginal Cost

Marginal cost (MC) is the cost of producing one extra unit and is calculated by dividing the change in total costs by change in output.

Revenue

- **Total Revenue (TR)** is the money the firm gets back from selling goods and is found by multiplying the number sold, Q, by the selling price, P.
- **Average Revenue (AR)** is the amount received from selling one item and equals the selling price of the good.
- **Marginal Revenue (MR)** is the additional revenue got when one more unit of the good is sold.

Equations

$$TC = VC + FC$$

$$VC = TC - FC$$

$$FC = TC - VC$$

$$AC = \frac{TC}{Q}$$

$$TR = P \times Q$$

$$AR = \frac{TR}{Q} = \frac{PQ}{Q} = P$$

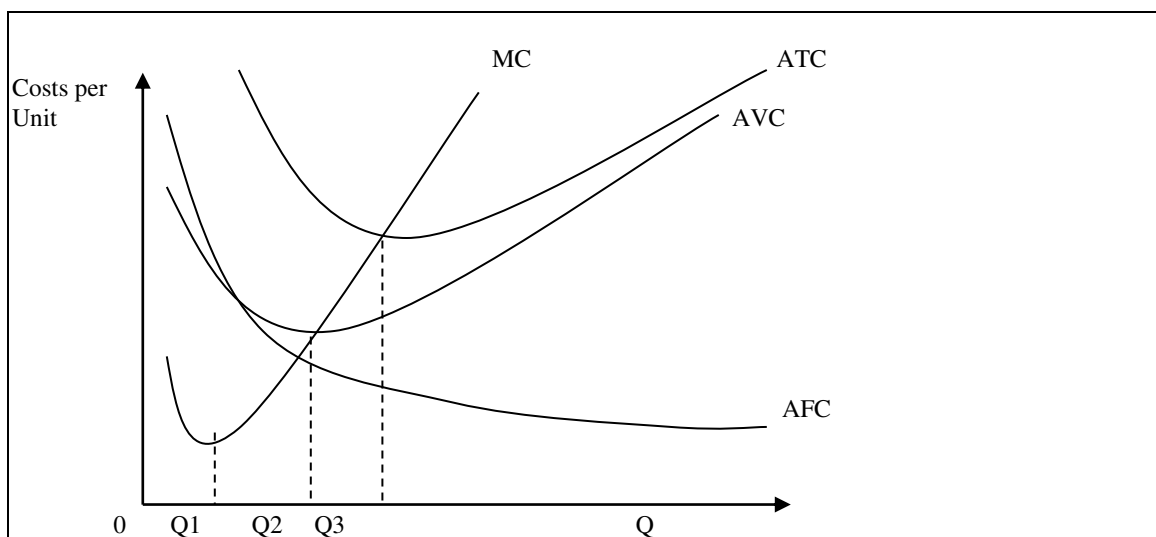
$$MC = \frac{\Delta TC}{\Delta Q}$$

$$MR = \frac{\Delta TR}{\Delta Q}$$

No of workers	Q	FC	VC	TC	AFC	AVC	AC	MC
0	0	500	0	500	---	---	---	
1	7	500	300	800	71.43	42.86	114.29	42.86
2	18	500	600	1100	27.78	33.33	61.11	27.27
3	33	500	900	1400	15.15	27.27	42.42	20.00
4	46	500	1200	1700	10.87	26.09	39.96	23.08
5	55	500	1500	2000	9.09	27.27	36.36	33.33
6	60	500	1800	2300	8.33	30.00	38.33	60.00
7	63	500	2100	2600	7.94	33.33	41.27	100.00
8	65	500	2400	2900	7.69	36.92	44.61	150.00
9	66	500	2700	3200	7.57	40.91	48.48	300.00
10	66	500	3000	3500				
11	64	500	3300	3800				
12	60	500	3600	4100				

If these figures are used the following is the diagram that you will get.

Per Unit Output Cost Curves



Social Cost

The private cost to a motorist of driving from Harare to Chitungwiza is the cost of petrol and oil and the wear and tear on his car. However, other people have to put up with the externalities of the journey, for instance the noise, smell, pollution and traffic congestion that the motorist helps to cause along the way.

If we add on to private cost an amount of money to compensate for the inconvenience caused, the overall figure will be the social cost of the journey:

Private costs + Externalities = Social cost

Cost to individual + Cost to other people = Cost to everyone

CHAPTER 5: MARKET STRUCTURES

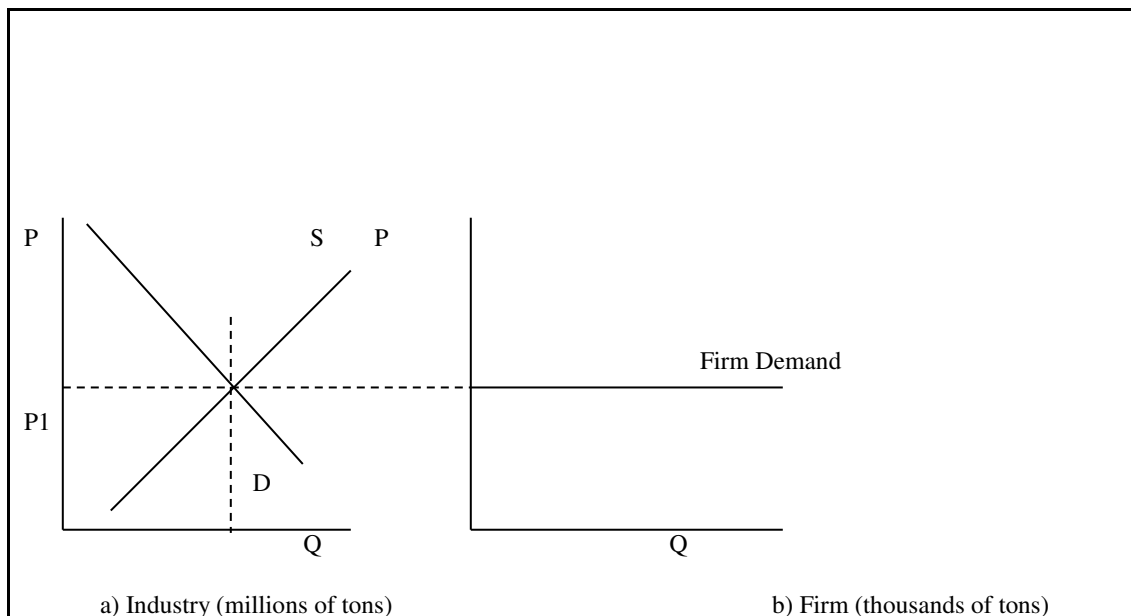
Below we are going to discuss four market structures, namely perfect competition, monopoly, monopolistic competition and oligopoly.

PERFECT COMPETITION.

Business firm - an organization set up and managed for the purpose of earning profits for its owner by producing goods and services for sale in markets.

Assumptions for Perfect Competition

- Firms are price takers. There are so many firms in the industry that each one produces an insignificantly small portion of total industry supply, and therefore has no power whatsoever to affect the price of the product. If a firm increases its price just slightly, then the quantity demanded of its product would drop to zero. It faces a horizontal demand curve at the market price: the price is determined by the interaction of demand and supply in the whole market.



The market price of eggs is P_1 . A competitive firm can sell all the eggs it wishes at that price. The output of any firm is a perfect substitute for that of any other firms. The market demand curve is downward sloping because consumers will buy more eggs at a lower price. The curve facing the firm is horizontal, because the firm's sales will have no effect on the price. $\square$

- There is complete freedom of entry of new firms into the industry. Existing firms are unable to stop new firms from setting business. Setting up a business takes time however. Freedom of entry, therefore applies in the long run. An extension of this assumption is that there is complete factor mobility in the long run. If profits are higher than elsewhere, capital will be freely attracted into that industry. Likewise if wages are higher than for equivalent work elsewhere, workers will freely move into that industry and will meet no barriers.
- All firms produce an identical product (the product is homogeneous). There is therefore no branding or advertising.
- No government intervention

- Producers and consumers have perfect knowledge of the market. That is the producers are fully aware of prices, costs and market opportunities. Consumers are fully aware of the price, quality and the availability of the product.
- Perfect mobility of the factors of production
- No entry/ exit barriers

The assumptions are very strict. Few if any industries in the real world meet these conditions. Certain agricultural markets are perhaps closest to perfect competition. The markets for fresh vegetables and grains (e.g. rapoko, maize, wheat. and mhunga) and also the market for bread.

Nevertheless despite lack of real world cases the model of perfect competition plays a very important role in economic analysis and policy. Its major relevance is its use as an ideal model.

The Short Run and the Long Run

In the short run the number of firms is fixed. Depending on its costs and revenue, a firm might be making large profits, small profits, no profits or loss and in the short run it may continue to do so.

In the long run, however, the level of profits will affect entry and exit from the industry. If the profits are high, new firms will be attracted into the industry, whereas if losses are being made firms will leave.

This leads to the distinction between normal and supernormal profits.

Normal Profit

This is the profit that is just enough to persuade firms to stay in the industry in the long run, but not high enough to attract new firms. If less than normal profits are made, firms will leave the industry in the LR.

Supernormal Profit

This is any above normal profit. If supernormal profits are made, new firms will be attracted into the industry in the long run. On the other hand if a firm makes losses in the long run some firms will leave the industry: and they will continue to do so until only normal profits are being made. Thus whether the industry expands or contracts in the long run will depend on the rate of profit.

The Short Run Equilibrium of the Firm

The determination of P, Q and Profit in the short run under perfect competition can best be shown in a diagram. Figure below shows SR equilibrium for both industry and a firm under perfect competition. Both parts of the diagram have the same scale for the vertical axis. The horizontal axes have totally different scales, however. For example if the horizontal axis for the firm were measured in say, thousands of tons, the horizontal axis for the whole industry might be measured in millions of tons.

Let us examine the determination of P, Q and profits in turn.

Price

The price is determined in the industry by the intersection of demand and supply. The firm faces a horizontal demand (AR) curve at this price. It can sell all it can produce at the market price (P_e), but nothing at a price above P_e .

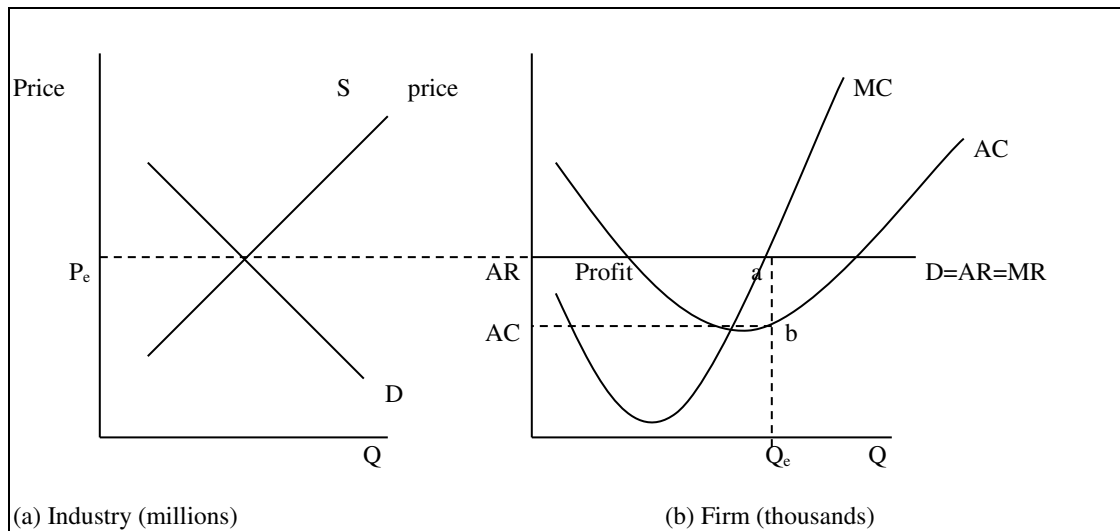
Output

The firms maximize profit where $MC=MR$, at output Q_e . Note that, since the price is not affected by the firm's output MR will equal the price.

$$TR = PQ$$

$$\frac{\Delta TR}{\Delta Q} = P$$

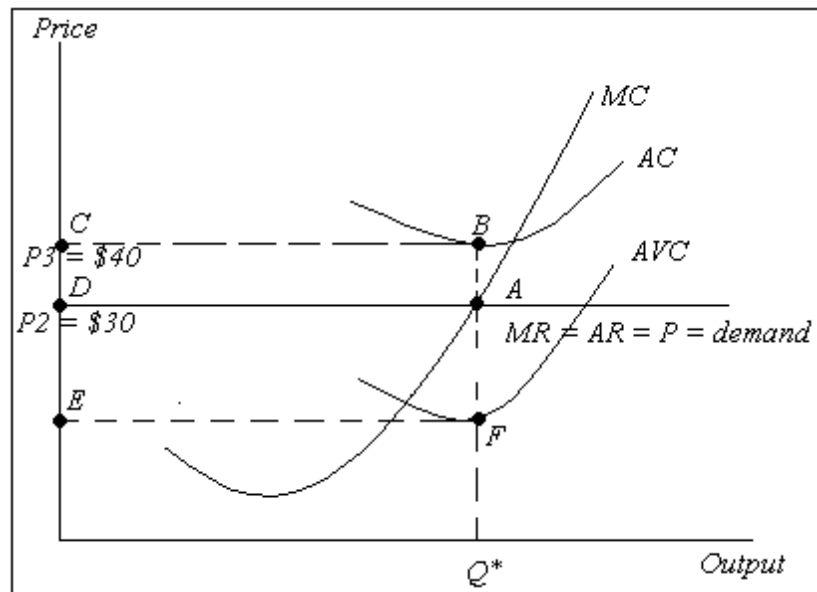
$$AR = \frac{TR}{Q} = \frac{PQ}{Q} = P$$



Total supernormal profit is given by area ARabAC.

What happens if the firm cannot make a profit at any level of output? This situation would occur if the AC curve were above the AR curve at all points. This is illustrated in figure below where the market price is P_1 . In this case, the point where $MC=MR$ represents the loss-minimizing point (where loss is defined as anything less than normal profit).

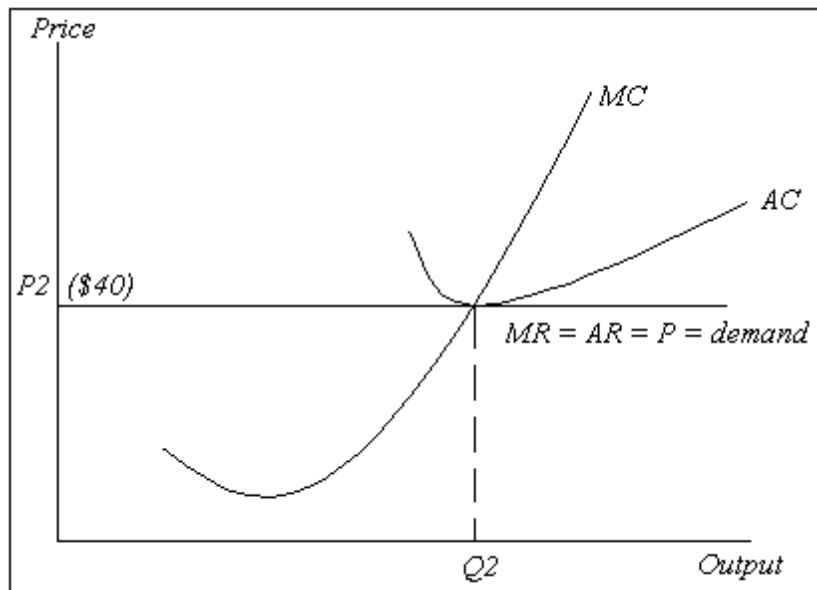
A competitive firm incurring losses.



At price $P_2 = \$30$ per unit the firm cannot avoid incurring losses, because that price is below the minimum average cost represented by $P_3 = \$40$. At the profit maximizing output Q^* the price P_2 is less than an average cost, so that line segment AB measures the average loss from production where $P = MC$, which represent, $AC - P$ (loss per unit). The firm could minimize its losses by producing at Q^* , with losses $ABCD$ being incurred. However if the firm were to shut down, it would incur even greater losses equal to the fixed costs of production $CBEF$

A COMPETITIVE FIRM BREAKING EVEN.

Normal Profit



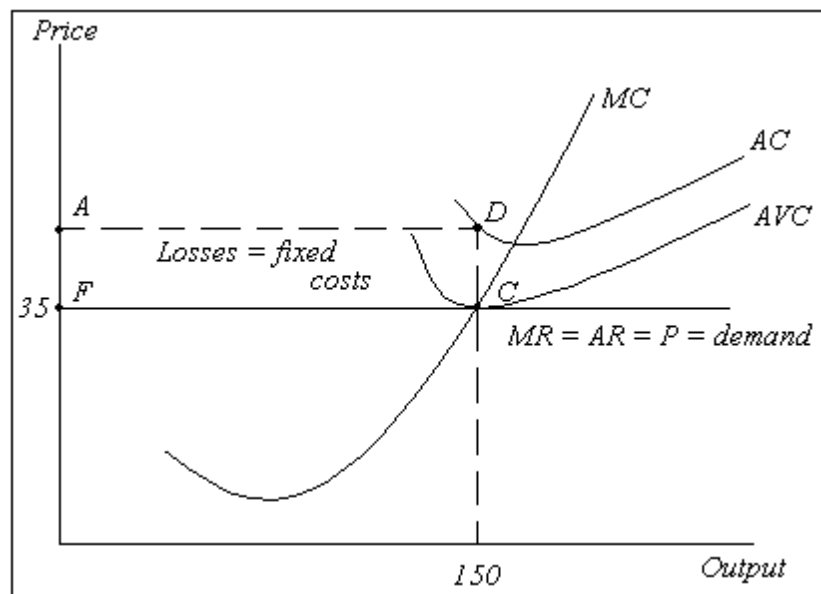
Firm is producing an output Q_2 , where $MC = MR$ at price (P_2) . Its total cost $(AC \times Q)$ equals its total revenue $(P \times Q)$. In this case the firm breaks even (or makes normal profit).

Shut Down Point.

Point at which the firm would be better off if it shuts down than it will be if it stays in business.

- If a price is above minimum (AVC) the firm will continue to produce in the short run.
- If a price is below minimum (AVC) the firm will shut down

The decision to shut down operations in the short run.

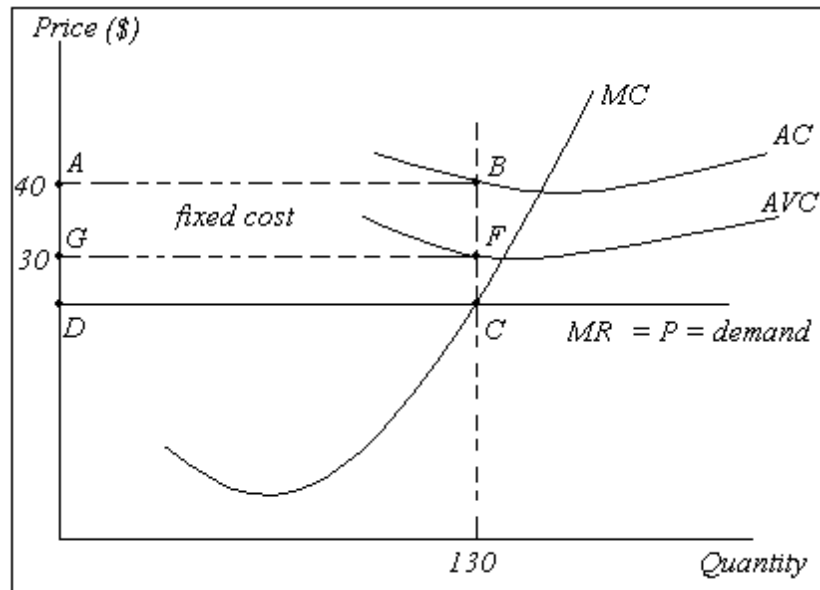


$$AFC = AC - AVC$$

$$\text{Per unit loss / profit} = P - AC$$

When price is falling to a level that just allows the firm its minimum possible average variable cost of input the firm is at shut down point. At a market price of \$35 per unit of that specific commodity $P = AVC$ at the output for which price is equal marginal cost ($P = MC$), the firm produces 150 units and a loss per unit is equal to the distance DC, which also represents the average fixed cost. Therefore at that output ($P - AC$) and ($AC - AVC$) are both equal to the distance DC. The economic losses incurred by continuing to operate are equal to fixed costs. If the price were to fall below \$35 per unit, the firm would close operations in the short run. Therefore the shut down point is at C, where the AVC curve is at minimum.

Shutting down.



At the output corresponding to the point at which $MR = MC$, operative losses represented by area ABCD exceed fixed cost, represented by area ABFG. Because the price is less than minimum AVC ($P < AVC \min$). This is because the vertical distance between the firm's demand curve and its average cost curve would exceed the vertical distance between AC and AVC. In this case the firm would be compelled to shut down operations immediately.

In summary, the conditions to remain in operation in the short run, while incurring losses are:

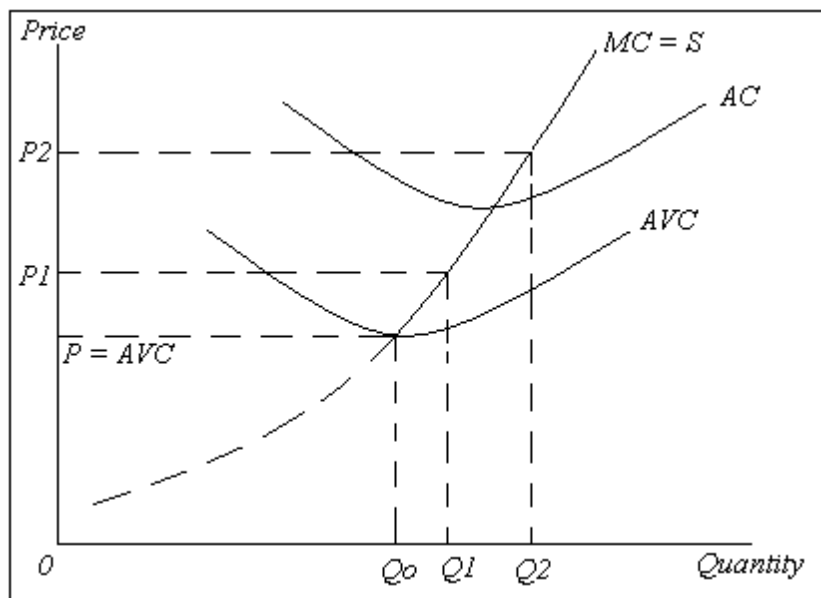
$$TR > VC$$

$$PQ > AVC(Q)$$

$$P > AVC$$

THE COMPETITIVE FIRM SHORT RUN SUPPLY - CURVE.

The competitive firm short-run supply curve is the portion of a competitive firm's marginal cost curve that lies above the minimum possible point of its AVC curve. Therefore, the MC curve gives the relationship between price and quantity supplied by the competitive firm. Short run supply curve slopes upwards because the firm MC tends to increase as output is increased. At any price below minimum possible AVC quantity supplied in the short run is zero. To determine the quantity supplied at any price greater than the minimum possible AVC, draw a horizontal line from that price to marginal cost curve. Dropping a vertical line to the horizontal axis gives the quantity supplied at that price.



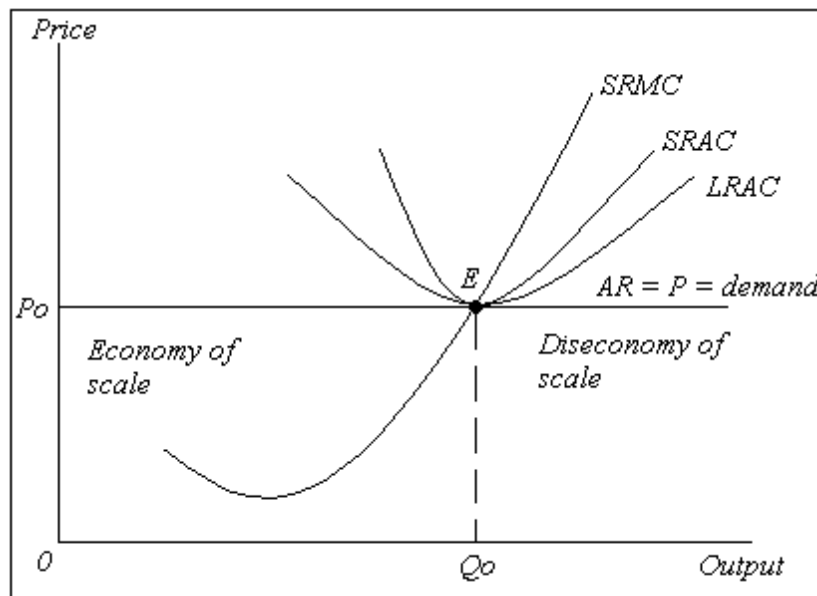
At a price P_1 quantity supplied is Q_1 . At a higher price P_2 quantity supplied is Q_2 . To induce the profit maximising firm to supply more output, market price should rise. When the market price increases, MR would exceed MC at current output. Therefore, the firm finds it profitable to increase quantity supplied at the new higher price and as a result increases MC to the point at which it equals that price.

The determinants of market supply.

- The number of firms in the industry
- The average size of firms in the industry measured by quantity of fixed inputs employed (for example average of factories for production capacity).
- The price of variable inputs used by firms in the industry.
- The technology employed in the industry.

LONG TERM EQUILIBRIUM UNDER PERFECT COMPETITION.

If firms are perfectly competitive and industry is making short-term surplus (profits), more firms will enter the industry. In the long run this will increase the market supply of the product and reduces the market price as well as the profits until all firms in the industry make a normal profit (break even). If on the other hand the firms are making losses firms will leave the industry and this will reduce supply and bid up the prices until only normal profits are made.



Economies of Scale are a long run phenomenon. Economies of scale are **internal** if they are Internal happening within the firm and **external** if they are happening at industry level.

Economies of Scale

These are economies made within a firm as a result of mass production. As the firm produces more and more goods, the average cost begin to fall because of:

- *Technical economies* made in the actual production of the good. For example, large firms can use expensive machinery, intensively.
- *Managerial economies* made in the administration of a large firm by splitting up management jobs and employing specialist accountants, salesmen, engineers etc.
- *Financial economies* made by borrowing money at lower rates of interest than smaller firms.
- *Marketing economies* These result from large companies making use of mass media e.g. television national press.
- *Commercial economies* made when buying supplies in bulk and therefore gaining a larger discount.
- *Research and development economies* made when developing new and better products.

External Economies of Scale

These are economies made outside the firm as a result of its location and occur when:

- A local skilled labor force is available.

- Specialist local back-up firms can supply parts or services.
- An area has a good transport network.
- An area has an excellent reputation for producing a particular good. For example, Bulawayo is associated with tyres in Zimbabwe

Internal Diseconomies of Scale

These occur when the firm has become too large and inefficient. As the firm increases production, eventually average costs begin to rise because:

- Management becomes out of touch with the shop floor and some machinery becomes over-manned.
- Decisions are not taken quickly and there is too much form filling.
- Lack of communication in a large firm means that management tasks sometimes get done twice.
- Poor labour relations may develop in large companies.

External Diseconomies of Scale

These occur when too many firms have located in one area. Unit costs begin to rise because:

- Local labour becomes scarce and firms now have to offer higher wages to attract new workers.
- Land and factories become scarce and rents begin to rise.
- Local roads become congested and so transport costs begin to rise.

In the diagram above long-term equilibrium occurs when price is P_0 and the business is operating at point E, any increase or decrease in output from point Q_0 would result in the firm making a loss.

Perfect Competition and Public Interest

There are a number of features of perfect competition that could be argued to be advantageous to society.

- Price equals MC. This is argued to be an optimal position. To demonstrate why consider the cases where they are not equal. At levels of output below equilibrium $P > MC$. This means that consumers put a higher value on consumption than it costs to produce additional output. It could be argued therefore, that more ought to be produced. If $P < MC$ it could be argued that less should be produced because consumers put a lower value on this additional output than it costs to produce. Clearly, if more should be produced when output is below equilibrium and less should be produced when output is above equilibrium, the equilibrium represents an optimum level of output. As we shall see later, it is only under perfect competition that P will equal MC .

- If a firm becomes less efficient than others, it will make less than normal profit and be driven out of business. If it is more efficient, it will earn supernormal profits (until other firms copy its more efficient methods). Thus competition between firms will act as a spur to efficiency.
- Similarly the desire for supernormal profit and the desire to avoid loss will encourage the development of new technology.
- There is no point in advertising under perfect competition, since all firms produce a homogeneous product (unless, of course, the firm believes that by advertising it can differentiate its product from its rivals' and thereby establish some market power, but then, by definition, the firm would cease to be perfectly competitive).
- LR equilibrium is at the bottom of the firm's LRAC curve. That is for any given technology, the firm, will produce at the least cost output in the long run.
- The consumer gains from low prices since not only are the costs kept low, but also there are no LR supernormal profits to add to these costs.

Limitations of Perfect Competition

At times perfect competition may be less desirable than other market structures such as control.

- Even though firms under perfect competition may seem to have an incentive to develop new technology they may not be able to afford the necessary research and development. Also, they may be afraid that if they did develop new more efficient methods of production, their rivals would merely copy them, in which case the investment would have been a waste of money.
- Perfectly competitive industries produce undifferentiated products. This lack of variety might be seen as a disadvantage to the consumer. Under monopolistic competition and oligopoly there is often intense competition over the quality and design of the product. This can lead to pressure on the firms to improve their products. This pressure will not exist under perfect competition.

IMPERFECT COMPETITION & MONOPOLY.

Imperfect competition.

Imperfect Competition prevails in a market whenever individual sellers have some degree of control (power) over the price of their output.

MONOPOLY

Monopoly is the market structure in which only one producer or seller exists for a product that has no close substitutes.

Characteristics of monopolies:

- There is only one firm which supply the entire market and many buyers and consumers
- The firm sells a unique product, which has no close substitutes.
- The firm has market power (that is it can control its price)
- Entry into the market is restricted, e.g. due to high costs and some special barriers to entry.

These barriers to entry are:

- High cost to enter a market that can support only one business, e.g. the supply of water and electricity etc.
- A business may have exclusive control of a natural resource. Other producers cannot compete, because they don't have that resource at their disposal. E.g. De Beers controls a large part of all diamond production, and this creates a barrier to entry for other firms.
- A business may have copyright or patent right on its product, thus making it illegal for other producer to duplicate the product.
- A monopoly may be created by the state making it legal.

- A well-known and popular trademark could ensure consumer loyalty, e.g. Pepsi.

Demand and marginal revenue.

Under pure monopoly, the business is the industry and faces the negatively sloped industry demand curve for its product. This means that if the monopolist wants to sell more of its product it must lower its price. Thus, for a monopolist MR is less than price, and the Marginal Revenue (MR) curve lies below the demand curve.

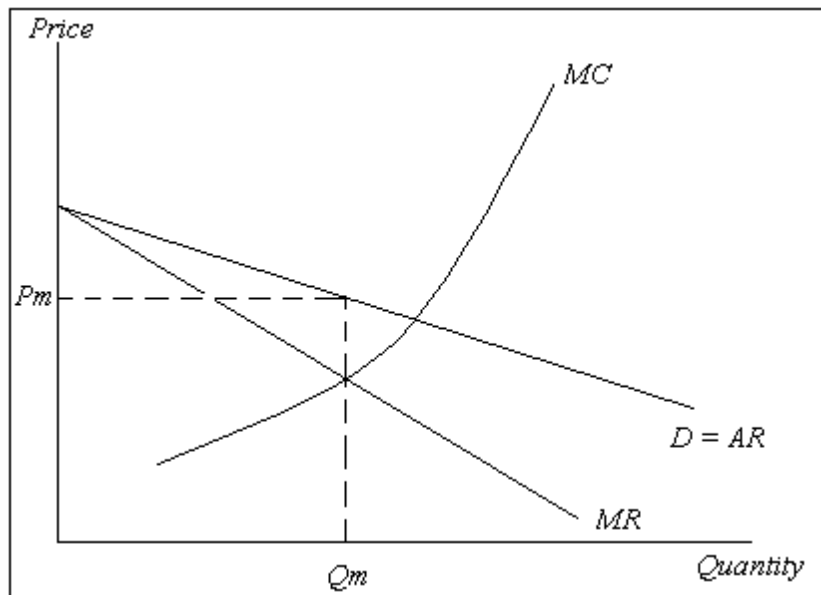
MONOPOLY: SHORT TERM EQUILIBRIUM.

Profit maximizing rule: Produce at an output level at which $MC = MR$.

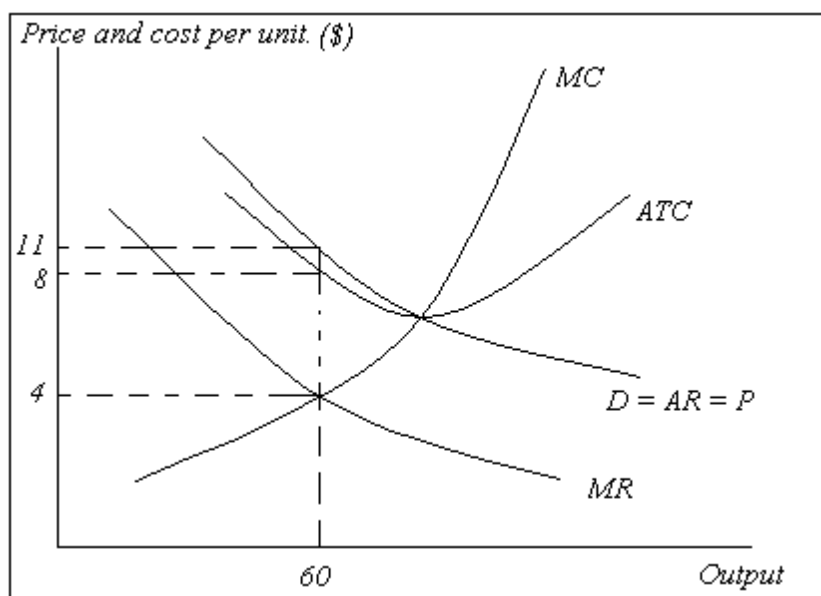
Finding the monopolist price and output.

Steps to follow:

- Find the profit maximizing output level where $MC = MR$.
- Extend the line up to the demand curve and down to the Q – axis, to determine the output Q_m , the monopoly chooses.
- From the point, where the line intersects with the demand curve, extend it horizontally to the P-axis or vertical axis. This will determine the price the monopolist will charge.



Consider the following diagram, which shows the demand and unit loss situation of a monopolist.



What is the output where the firm's profits are maximized? - 60.

The price at that output level is - \$11.

The average total cost at that output level is - \$8.

The profit / loss per unit is: $P - ATC = \$11 - \$8 = \$3$.

The total revenue at this output is: $TR = P \times Q = \$11 \times 60 = \660 .

The total cost at that output level is: $TC = TVC + TFC = ATC \times Q = \$8 \times 60 = \$480$

The total profit / loss at this output level is: $T\pi = TR - TC = \$660 - \$480 = \$180$.

Where © = Profit.

MONOPOLISTIC COMPETITION

This was a theory developed in the 1930's by the American economist Edward Chamberlain. Monopolistic competition is nearer to the competitive end of the spectrum. It can best be understood as a situation where there are a lot of firms competing, but each firm does nevertheless have some degree of market power (hence the term monopolistic competition); each firm has some discretion as to what price to charge for its products.

Assumptions of monopolistic competition

- There are quite a number of firms. As a result, each firm has an insignificantly small share of the market, and therefore its actions are unlikely to affect its rivals to any great extent. What this means is that each firm in making its decisions does not have to worry how its rivals will react. This assumes that what its rivals choose to do will not be influenced by what it does. This is known as the assumption of independence (as we shall see later this is not the case under oligopoly). There we assume that firms believe that their decisions do affect their rivals, and that their rivals' decisions do affect them. Under oligopoly we assume that firms are interdependent).
- There is freedom of entry of new firms into the industry. If any firm wants to set up business in this market, it is free to do so. In these two respects, therefore, monopolistic competition is like perfect competition.

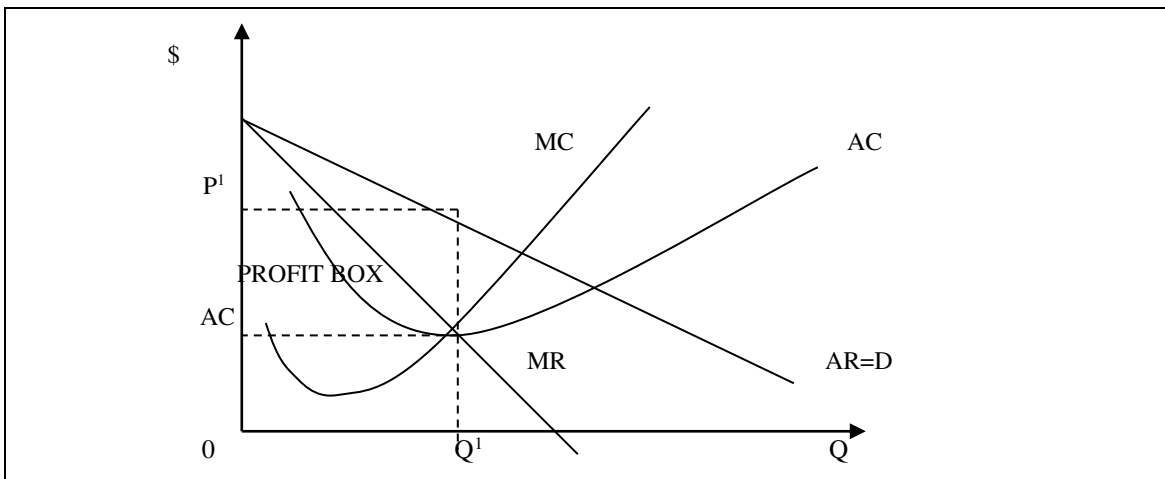
- c) Unlike perfect competition, however, each firm produces a product or provides a service in some way different from its rivals. The firm has monopoly over its brand, but faces competition in the overall product range. As a result, it can raise its price without losing its customers. Thus the curve is downward sloping, albeit relatively elastic given the large number of competitors to whom customers can turn. This is known as the assumption of **product differentiation**.

Petrol stations, restaurants, hairdressers & builders are all examples of monopolistic competition.

SR EQUILIBRIUM OF THE FIRM

As with other market structures, profits are maximized at $MC = MR$. The diagram will be the same as for the monopolist except that the AR & MR curves will be more elastic. This is illustrated in the figure below. As with competition, it is possible for the monopolistically competitive firm to make supernormal profit in the SR. This is shown in the diagram below. Just how much profit the firm will make in the SR depends on the strength of demand: the position & elasticity of demand. The further to the right the demand curve is relative to the average cost curve, and the less elastic the demand curve is, the greater will be the firm's profit. Thus a firm facing little competition and whose product is considerably differentiated from its rivals may be able to earn considerable SR profits.

SR equilibrium under monopolistic competition

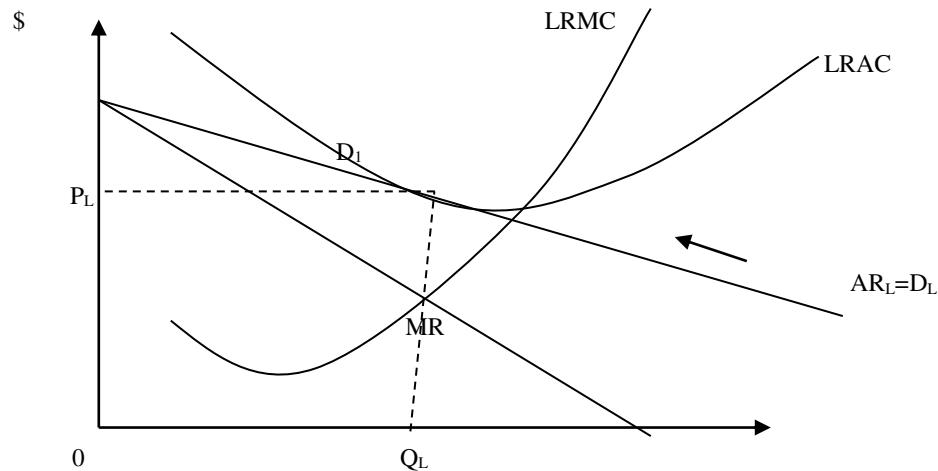


LONG RUN EQUILIBRIUM

If typical firms are earning supernormal profit, new firms will enter the industry in the LR. As new firms enter, they will take some of the customers away from the existing firms. The demand for existing firms will therefore fall. Their demand (AR) curve will shift to the left & will continue doing so as long as supernormal profits remain and thus new firms continue entering. LR equilibrium will be reached when only normal profits remain: when there is no further incentive for new firms to enter. This is illustrated in figure below.

The firm's demand curve settles at D_L , where it is tangential to the firm LRAC curve. Output will be Q_L : where $AR_L = LRAC$. At any other output, LRAC is greater than AR thus less than normal profit would be made.

LR equilibrium of the firm under monopolistic competition



The firm under monopolistic competition does not achieve allocative efficiency and productive efficiency. Furthermore there is excess capacity, which is shown by the difference between Q_L and the output it would otherwise produce, were it operating at minimum LRAC.

LIMITATIONS OF THE MODEL

There are various problems in applying the model of monopolistic competition to the real world:

- Information may be imperfect. Firms will not enter an industry if they are unaware of the supernormal profits currently being made or if they underestimate the demand for the particular product they are considering selling.
- Given that the firms in the industry produce different products, it is difficult if not impossible to derive a demand curve for the industry as a whole. Thus the analysis has to be confined to the level of the firm.
- Firms are likely to differ from each other not only in the product they produce or the service they offer, but also in their size and in their cost structure. What is more, entry may not be completely unrestricted. Two petrol stations could not set up in exactly the same place – on busy crossroads. Thus although the typical or representative firm may earn only normal profit. They may have the cost advantage or produce a product that is impossible to duplicate perfectly.

One of the biggest problems with the simple outlines in the previous sections is that they concentrate on price & output decisions. In practice, the profit maximising firm under monopolistic competition will also need to decide the exact variety of product to produce & how much to spend on advertising it. This will lead the firm to take part in non-price competition.

NON – PRICE COMPETITION

Non – price competition involves two major elements: product development and advertising. The major aims of product development are to produce a product that will sell well (i.e. one in high or potentially high demand) & that is different from rivals' products (i.e. has a relatively inelastic demand due to lack of close substitutes). In the case of shops or other firms providing a service, product development will take the form of attempting to provide a service which is better than, or at least different from, that of rivals: personal service, late opening, certain lines stocked etc.

The major aim of advertising is to sell the product. This can be achieved not only by informing the consumer of the product's existence and availability, but also by deliberately trying to persuade customers to purchase the goods. Like product development, successful advertising will not only increase demand, but also makes the firm's demand curve less elastic since it stresses the specific qualities of the firm's product over its rivals.

Product development and advertising not only increase a firm's demand and hence revenue, they involve increased costs. So by how much should firm advertise, to maximise profits?

For any given price & product, the optimal amount of advertising is where the revenue from additional advertising (MR_A) is equal to its cost (MC_A). As long as $MR_A > MC_A$ additional advertising will add to profit. But extra amounts spent on advertising are likely to lead to smaller & smaller increases in sales. Thus MR_A falls, until $MR_A = MC_A$.

At that point no further profit can be made. It is a maximum.

Two problems with this analysis:

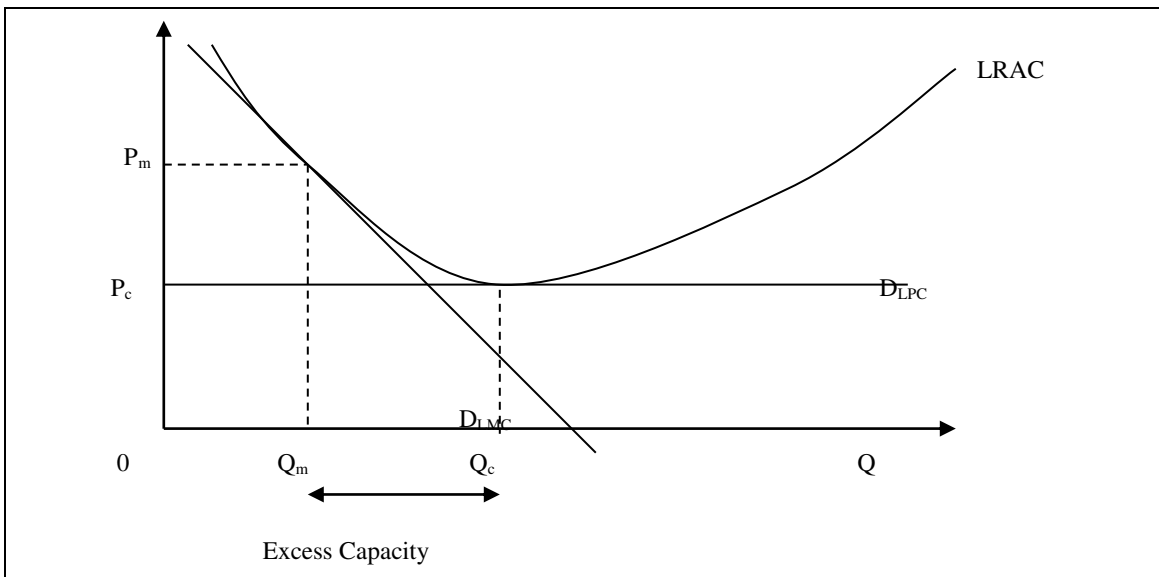
- The effect of product development and advertising on demand will be difficult for a firm to forecast.
- Product development and advertising are likely to have different effects at different prices. Profit maximisation, therefore, will involve the more complex choice of the optimum combination of the price, type of product and the level and variety of advertising.

MONOPOLISTIC COMPETITION & THE PUBLIC INTEREST

Comparison with perfect competition

It is often argued that monopolistic competition leads to less efficient allocation of resources than perfect competition. Figure below compares the LR positions for two firms. One is under perfect competition and thus faces a horizontal demand curve. It will produce an output of Q_c at a price of P_c . The other is under monopolistic competition and thus faces a downward sloping demand curve. It will produce the lower output of Q_m at a higher price of P_m .

LR equilibrium of the firm under perfect & monopolistic competition



Where D_{LPC} is the long run demand for a firm in perfect competition and D_{LMC} is the long run demand curve for a firm under monopolistic competition.

A crucial assumption here is that the firms would have the same LR average costs (LRAC) curve in both cases. Given this assumption monopolistic competition has the following disadvantages:

- a) Less will be sold at a higher price
- b) Firms will not produce at the least cost point.

By producing more, firms would move to a lower point on their LRAC. Thus, firms under monopolistic competition are said to have excess capacity. In figure above this excess capacity is shown as $Q_1 - Q_2$.

COMPARISON WITH MONOPOLY

The arguments here are very similar to those when comparing perfect competition and monopoly. On the other hand freedom of entry for new firms and hence lack of LR supernormal profit under monopolistic competition are likely to help keep prices down for the consumer and encourage cost saving. On the other hand monopolies are likely to achieve greater economies of scale and have more funds for R & D investment.

Furthermore, the consumer may benefit from monopolistic competition by having a greater variety of products to choose from. Each firm may satisfy some particular requirement of particular consumers.

OLIGOPOLY: THE THEORY OF FEW SELLERS

Monopoly is a theory of a market with one seller and many buyers. Supply and demand is the theory of a market with many sellers and many buyers. What changes in these theories would be needed to tell us what happens if there is **oligopoly**, that is, more than one seller, but fewer than many? What, for example, happens if there are two sellers, or **duopoly**? Though many economists suspect that the results of two sellers are more similar to those of one seller than to those of many sellers, economics lacks a clear theory that can prove this suspicion.

The problem of **interdependence** has thwarted economists' attempts to develop a good theory of oligopoly. When there are only a few sellers, each recognizes that his decisions affect others who may react to what he does.

When the possession of market power is profitable, it should attract new entrants into the industry. If **entry** is easy, then the existence of very few or even only one firm may not result in economic inefficiency. The threat of potential entry may be enough competition to keep the industry operating at or close to the competitive solution. In this case, the market is a **contestable market**. However, if entry is not easy but there are significant **barriers to entry**, the threat of competition is less. Barriers to entry exist when there are **sunk costs**--expenses that cannot be recovered once a firm has entered the industry. Where these costs are high, the industry probably operates as the theory of monopoly suggests it will.

Barriers to entry can take several forms. They will exist if large amounts of specialized machinery are required to enter an industry and resale of that machinery is difficult. They

will exist if a firm must establish a reputation for the quality or reliability of a product. They will also exist if a firm must expend resources in order to get governmental approval to enter. In all of these cases, the barriers to entry can be viewed as sunk costs.²

Collusive oligopoly.

In contrast, there are times when great numbers of sellers are able to organize and act as a unified seller. Sellers have the incentive to act in this way because it will increase profits. This form of oligopoly is called collusive oligopoly and the unified organisation that results is called a cartel. The profit maximisation under collusive oligopoly is the same as the one for monopoly, since this effectively becomes the only seller in the market.

Conditions under which cartels are likely to Flourish

Cartel are likely to last longer in an environment with the following features:

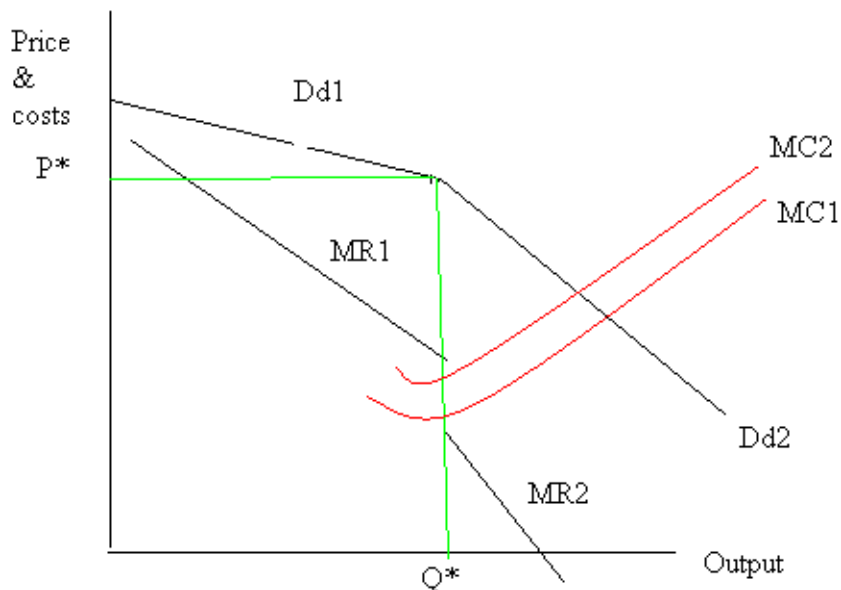
- There should be few firms so that coordinating their decisions becomes easier.
- The firms should have similar cost structures and technology of production so that a common, mutually acceptable price can be a reality.
- There should not be cheating by the members of the cartel, so that the firms observe or adhere to their allocated production quotas to avoid over-production.
- The legal framework should not be too restrictive, otherwise the cartels will be outlawed.
- The macroeconomic environment should be relatively stable so the firms do not need to regularly meet and agree on a new price, each time creating the possibility of the cartel collapsing through lack of consensus.

Non-collusive Oligopoly.

is when the firms decide to act independently in the oligopoly market. This creates a lot of uncertainty in the market. The result is price undercutting and advertising wars as firms try to outdo each other. The general assumption in the analysis of non-collusive oligopoly:

- If a firm raises the price of its product, other do not follow, and as a result it loses sales.
- When the firm reduces its price others follow suit, i.e they match the price reduction and therefore there is no gain in sales to be expected.

This implies that rivals are more responsive to price reductions than they are to price increases initiated by the firm. This results in a kinked demand curve, a demand curve with two different slopes, one elastic and the other inelastic.



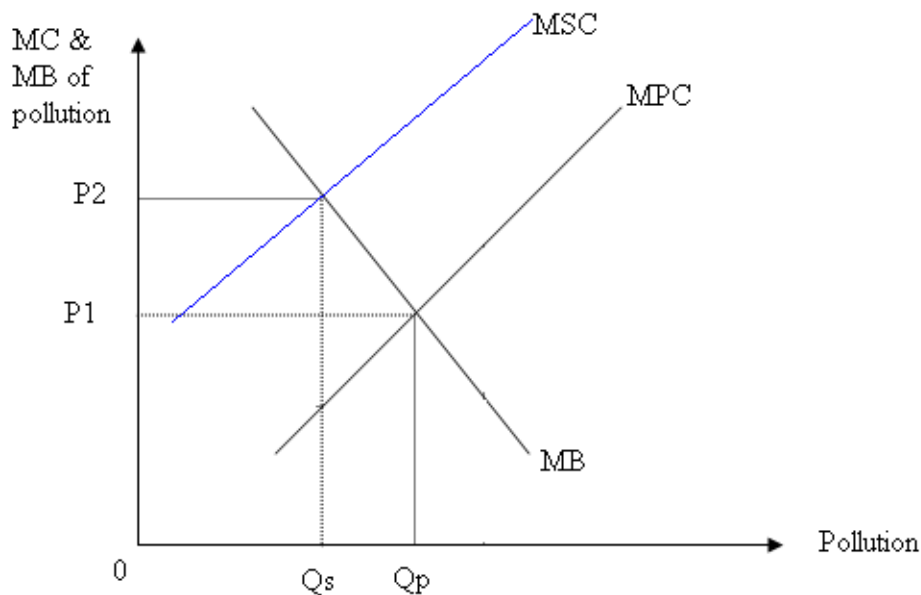
From the above graph, Dd1 is the elastic part of the demand curve, with MR1 as its accompanying marginal revenue curve. Similarly, Dd2 is the elastic part of the demand curve with MR2 as the marginal revenue curve. Applying the profit maximization rule implies that the level of output that gives maximum profit will be Q^* and the optimal price at P^* . The price is rigid or sticky at P^* over a long time as no firm is willing to initiate price changes. From the firm's perspective, it is not wise to increase or reduce price as there are always detrimental effects both ways. If for instance the costs increase from MC1 to MC2, the firm is forced to absorb the cost increase in an attempt to avoid price adjustments. The price will only change if the costs increase beyond the 'jump' in marginal revenue.

CHAPTER 6: EXTERNALITIES

Definition of Externality

An externality is a situation where the actions of one's production or consumption activities affects other economic agents who are not directly involved in the production or consumption of the commodity. An externality is also referred to as a 'spillover effect' or 'third party effect' because it affects third parties. When a smoker decides to consume

cigars in public, the non-smokers nearby may be harmed by his actions, and this can be considered as a negative externality since it is detrimental to third parties. Similarly if Collen Bawn cement factory produces pollutants in the process of producing cement, and eventually cause environmental damage affecting the Gwanda communities, then it is generating a negative externality. Thus a negative externality is one which inflicts harm or damage to third parties. When there



is a negative externality in production, the firm's private costs will diverge from the social costs since the social costs would include the pollution, which is a negative externality:

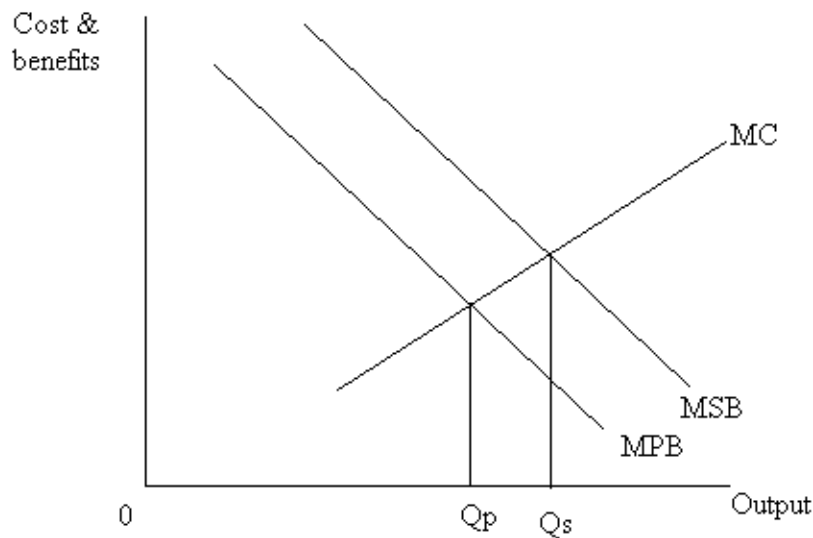
$$\text{Marginal social cost} = \text{marginal private cost} + \text{external cost}$$

$$MSC = MPC + E$$

The external cost is the cost associated with pollution. The firm considers the private costs that it directly incurs and totally disregards the external cost of pollution. The ideal output from the firm's perspective then is Q_p . Society, however, considers both the private cost to the firm and the external cost imposed by pollution. As such the marginal social cost is larger than the marginal private cost as shown in the above graph. The optimal output from society's perspective is Q_s .

The positive externalities

A positive externality is a situation where the actions of one economic agent create benefits to other economic agents who did not contribute towards the economic activity. If, for instance, I hire a security company to protect my premises, my neighbours may also benefit as the chances of robbery within the neighbourhood may be significantly reduced, thus creating a positive externality.



From the diagram above, Q_p is the output that is ideal from the perspective of private decision-makers since they would only consider the marginal private benefit. However, society would consider the marginal social benefit and prefers output of Q_s .

If a commodity has positive externality, the social benefit will diverge from the private benefit:

$$\text{Marginal social benefit} = \text{marginal private benefit} + \text{external benefit}$$

$$MSB = MPB + E$$

As such the two can only be made to converge if the government introduces a subsidy to promote the production of the commodity.

Solutions to the Externality Problem

The externality problem can be solved or eliminated through the use of any of the following:

1. **Command and Control:** It involves initiating laws or regulations that ban the generation of negative externalities. Some of the mechanisms might involve not only targeting the externality but also targeting the activity caused by the externality. There are adverse effects if such measures are implemented, for instance, it would be physically impossible for most manufacturing processes no matter how beneficial their output is, to remain operational without generating any form of pollution.
2. **Pigouvian Tax:** It suggests the use of taxes to achieve efficiency in the presence of negative externalities and subsidies in the presence of positive externalities. Pigou argued that when competition rules with social and private net products at the margin diverging, then it is theoretically possible to put matters right by imposition of a tax or granting a subsidy. The polluting firm should be taxed according to the amount of the pollutant emitted. If the firm is producing output X and selling at per unit price P_x and emitting S amount of pollutant, then its profit maximization problem can be stated as:

$$\text{Max}\pi = P_x X - C_x(X; S)$$

Where: C_x is the cost of production

If a Pigouvian tax of t per unit of pollution is introduced, then the profit maximization problem becomes:

$$\text{Max}\pi = P_x X - C_x(X; S) - tS$$

3. Pollution Quota

This quantity restriction on how much pollutant each firm is allowed to produce. The firm should improve their production to ensure that they do not exceed the allocated quota. There are always costs associated with reducing the pollution levels. For instance firms have to make investment in technology that is compatible with the permissible pollution levels. There are also penalties for violating the pollution quota. In an economy where there are two firms 1 and 2; the quota for firm 1 can be

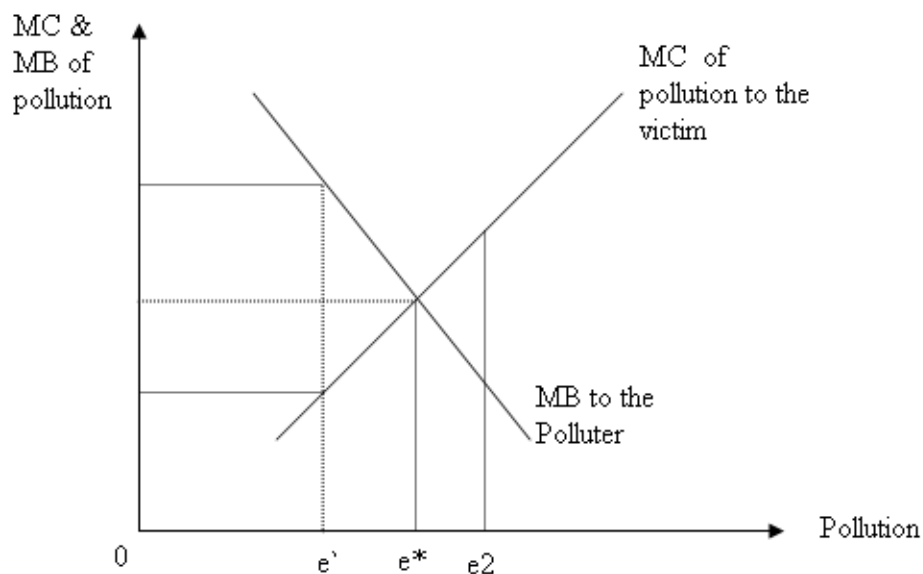
represented by X_1 and the quota for firm 2 is represented by X_2 . The target maximum emission level, that is the aggregate pollution level expected when both firms observe the pollution quota is equal to:

$$X = X_1 + X_2$$

4. Allocating Property Rights

The use of property rights is based on Coase Theorem. The theorem says if property rights are complete and if parties can negotiate an efficient solution to the externality; if private parties can bargain without transaction costs over the allocation of resources; then such bargaining will always solve problems of externalities on its own and allocate resources efficiently regardless of who owns the property rights. According to this theorem, an externality is an outcome of the absence of complete property rights, that is the existence of common property, which normally leads to what is referred to as 'Tragedy of the Commons'. The 'Tragedy of the Commons' implies that common property is in danger of being overused with the risk that it will eventually be depleted or dilapidated beyond any possible reclamation.

If the firm has the property rights, then the environment becomes its property and it will have an interest in preventing its deterioration. Thus there will be an incentive to minimize pollution levels. Similarly, if the community has the property rights from the polluting firm, it can claim compensation as it is entitled to do so. Whoever has the property rights has the chance to generate revenue from pollution. The allocation of property rights may therefore have a redistribution effect. The above can be graphically represented as shown below:



The optimal pollution level is where marginal benefit (MB) to the polluter is equal to the marginal damage to the victim. If firm 1 has the property rights, it can allow firm 2 to increase the pollution level from e' to e^* . At e' marginal benefit to the polluter is greater than the marginal damage to the victim. This is because at e' firm 2 enjoys a greater marginal benefit from polluting than the marginal damage endured by firm 1. Therefore firm 2 can compensate firm 1. If firm 2 (polluter) has the property rights, then it can emit pollution level e_2 unless it is compensated or bribed for reducing pollution levels. Firm 1 will face pollution level e_2 unless it accepts to pay the bribe for pollution reduction to firm 2. Firm 1 will face a marginal damage which is greater than the marginal benefit to the polluter.